

A Survey on Model-Based Distributed Control and Filtering for Industrial Cyber-Physical Systems

Derui Ding , Member, IEEE, Qing-Long Han , Fellow, IEEE, Zidong Wang , Fellow, IEEE, and Xiaohua Ge , Member, IEEE

Abstract—Industrial cyber-physical systems (CPSs) are large-scale, geographically dispersed, and life-critical systems, in which lots of sensors and actuators are embedded and networked together to facilitate real-time monitoring and closed-loop control. Their intrinsic features in geographic space and resources put forward to urgent requirements of reliability and scalability for designed filtering or control schemes. This paper presents a review of the state-of-the-art of distributed filtering and control of industrial CPSs described by differential dynamics models. Special attention is paid to sensor networks, manipulators, and power systems. For real-time monitoring, some typical Kalman-based distributed algorithms are summarized and their performances on calculation burden and communication burden, as well as scalability, are discussed in depth. Then, the characteristics of non-Kalman cases are further disclosed in light of constructed filter structures. Furthermore, the latest development is surveyed for distributed cooperative control of mobile manipulators and distributed model predictive control in industrial automation systems. By resorting to droop characteristics, representative distributed control strategies classified by controller structures are systematically summarized for power systems with the requirements of power sharing and voltage and frequency regulation. In addition, distributed security control of industrial CPSs is reviewed when cyber-attacks are taken into consideration. Finally, some challenges are raised to guide the future research.

Index Terms—Distributed control, distributed filtering, droop characteristics, industrial cyber-physical systems (CPSs), power schedule.

Manuscript received November 12, 2018; revised February 10, 2019; accepted March 12, 2019. Date of publication March 15, 2019; date of current version May 2, 2019. This work was supported in part by the Australian Research Council Discovery Project under Grant DP160103567, in part by the National Natural Science Foundation of China under Grant 61573246, and in part by the Natural Science Foundation of Shanghai under Grant 18ZR1427000. Paper no. TII-18-3010. (Corresponding author: Qing-Long Han.)

D. Ding, Q.-L. Han, and X. Ge are with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC 3122, Australia (e-mail: deruiding2010@usst.edu.cn; qhan@swin.edu.au; xge@swin.edu.au).

Z. Wang is with the Department of Computer Science, Brunel University London, Uxbridge UB8 3PH, U.K. (e-mail: Zidong.Wang@brunel.ac.uk).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TII.2019.2905295

I. INDUSTRIAL CYBER-PHYSICAL SYSTEMS

A CYBER-physical system (CPS), a fast growing research area, is a highly integrated system of physical components involving sensors, actuators, and various equipments, as well as cyber, possessing ubiquitous computation and efficient communication. From the engineering point of view, CPSs are considered as the most promising industrial systems including transportation networks, energy systems, water/gas distribution networks, and unmanned factories. For example, multiple industrial robots with an inertial navigation device or various sensors are programmed for the movement along a programmed trajectory to cooperatively complete production tasks [1], [2]. The main advantage of these systems is that the tight coordination of cyber and physical elements provides greater autonomy, efficiency, functionality, reliability, and adaptability. Furthermore, industrial CPSs are regarded as a core ingredient in the so-called fourth industrial revolution [3], [4], and lots of efforts are made to establish their important position, such as Industry 4.0 in Germany and Industrial Internet in the USA.

Industrial CPSs are large-scale, geographically dispersed, federated, cooperative, and life-critical systems, in which lots of sensors and actuators are embedded and networked together to facilitate real-time monitoring and closed-loop control. In addition, the implementation of industrial CPSs largely depends on sensor networks and distributed control networks. In other words, these two networks are usually indispensable in a largely deployed CPS architecture [5]. Specifically, sensor networks are usually deployed in the interior or neighboring of plants to gather various critical information with the purpose of making correct perception of the physical plants. With the help of these information, actuators can have the opportunity to do real-time reaction to some changes of physical plants. As such, a closed loop is formed when integrating cyber and physical worlds through both communicating between sensors and actuators and operating plants via actuators [6]. In the past few years, many of commercial sensors and actuators equipped with some communication and data processing capabilities are made available benefiting from the hard work of research institutions and companies from around the world.

As is done for almost all of real-world engineering systems, the model-based performance analysis of an industrial CPS plays an important role in understanding and adjusting

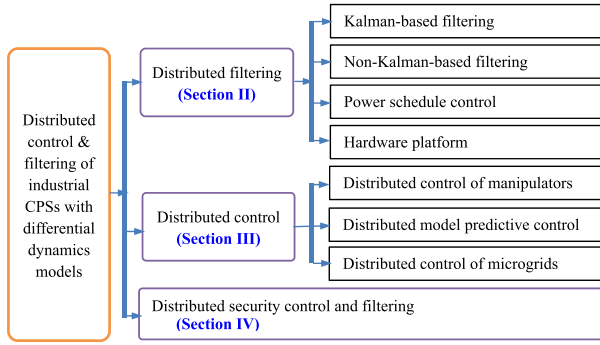


Fig. 1. Structure of this survey.

its dynamic behavior. Because of the limitation of geographic space and various resources in energy and communication aspects, there is an ever-increasing need to execute: distributed control for satisfying reliability and scalability requirements while maintaining stability performance; and distributed filtering for achieving scalability and disturbance attenuation capability while ensuring expected filtering accuracy. However, traditional tools developed in the central paradigm are not feasible to meet the demands of industrial CPSs mainly due to their inherent dynamic nature. Specifically, on the one hand, the complexity of industrial CPSs, in terms of its degree or intensity, could be great enhanced due to various network-induced phenomena and employed communication protocols [7], [8]. On the other hand, the varying topology, coming from connection failures or cyber-attacks [9], gives rise to an obstacle of the application of developed analysis approaches. As such, it is of both theoretical significance and practical importance to develop an effective framework of performance analysis and synthesis under the distributed paradigm.

It is thus desirable to survey what results have been developed in the field of industrial CPSs modeled by differential dynamics equations and further identify what challenges need to be dealt with. For this purpose, this paper intends to provide a review of the state-of-the-art of distributed control and filtering of industrial CPSs on sensor networks, manipulators, and power systems (see Fig. 1 for the organization). Specifically, for real-time monitoring, we identify key techniques of Kalman-based filtering algorithms, review typical filter structures utilized in a non-Kalman framework, and provide a systematic performance analysis from three aspects: calculation burden, communication burden, and scalability. For distributed closed-loop control, we first survey the latest advances on distributed control of manipulators in industrial automation systems and then summarize typical distributed control strategies in microgrids via droop characteristics with the requirements of power sharing and voltage and frequency regulation. Furthermore, we summarize recent developments on distributed security control of industrial CPSs when a cyber-attack is a concern. Finally, we raise some challenging issues to guide the future research.

II. DISTRIBUTED FILTERING FOR INDUSTRIAL CPSs

As an indispensable part, sensor networks are usually seamlessly integrated into industrial CPSs to facilitate the real-time

sensing, monitoring, and control. Sensor nodes in reality are sparsely deployed in a predetermined field along with subsystems or distributed actuators. Benefiting from embedded algorithms, i.e., distributed filtering algorithms, these sensor nodes can effectively process the collected information in function and meanwhile exchange necessary data via a communication topology. As such, distributed filtering based on sensor networks fills in the gap between the wealth of distributed information and the understanding of physical behavior [10]. Therefore, it receives considerable research interest in the past few years and some typical filtering algorithms are proposed in the literature. It is worth noting that, in comparison with the centre filtering or fusion, a critical challenge here is how to ensure the consensus or global performance of all estimates while guaranteeing the effectiveness of filtering algorithms. Various consensus strategies are proposed with intent to deal with such a challenge. Up to date, two fashionable schemes realizing both consensus and effectiveness of filtering are, respectively, the iterative computation via captured information and the design of distributed filtering structures via distributed information fusion. In this section, the latest development is systematically surveyed and some typical applications in industrial CPSs are further summarized.

A. Kalman-Based Distributed Filtering

In order to clearly present the filtering structure and discover the corresponding performance, let us introduce the following simple dynamics process:

$$\begin{cases} x_{k+1} = A_k x_k + w_k \\ y_{i,k} = C_{i,k} x_k + v_{i,k} \end{cases}$$

for $i \in \{1, 2, \dots, N\}$, where $x_k \in \mathbb{R}^n$ is the process state and $y_{i,k} \in \mathbb{R}^m$ is the measurement on the i th sensor. The disturbances $w_k \in \mathbb{R}^n$ and $v_{i,k} \in \mathbb{R}^m$ are mutually independent white Gaussian random variables with zero mean values and bounded covariance $Q > 0$ and $R_i > 0$.

Define that the *local* and *modified* updates are, respectively, $\hat{x}_{i,k}^o$ and $\hat{x}_{i,k}$, and the one-step prediction is $\hat{x}_{i,k}^-$. It is well known that the local Kalman filtering on the i th sensor can be expressed as

$$\begin{cases} \hat{x}_{i,k}^- = A_k \hat{x}_{i,k-1} \\ \hat{x}_{i,k}^o = \hat{x}_{i,k}^- + K_{i,k} (y_{i,k} - C_{i,k} \hat{x}_{i,k}^-) \\ P_{i,k}^- = A_k P_{i,k-1} A_k^T + Q \\ K_{i,k} = P_{i,k}^- C_{i,k}^T (C_{i,k} P_{i,k}^- C_{i,k}^T + R_i)^{-1} \end{cases} \quad (1)$$

where $P_{i,k}^-$ and $P_{i,k}$ are the prediction and estimation error covariances and $K_{i,k}$ is the optimal filter gain under the given performance index.

In this survey paper, the topology of sensor networks is described by a triple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{S})$. In this triple, $\mathcal{V} = \{1, 2, \dots, N\}$, $\mathcal{E}$, and $\mathcal{S} = [\alpha_{ij}]_{N \times N}$ with nonnegative adjacency element α_{ij} stand for the sets of nodes, edges, and adjacency matrix, respectively. The set of neighbors of the i th node is denoted by $N_i = \{j : (i, j) \in \mathcal{E}\}$ and Θ_i stands for

Algorithm 1

- 1: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$, $\hat{x}_{i,k}^o$, $P_{i,k}^-$ and $K_{i,k}$ via (1);
- 2: *Consensus calculation*
- 3: Initialize $\Omega_{i,k}(0)$, $q_{i,k}(0)$ and the step length L ;
- 4: Repeat each step until $l = L - 1$;
- 5: *Communications*
Broadcast $\Omega_{i,k}(l)$ to all neighbors;
Receive $\Omega_{j,k}(l)$ from all neighbors $j \in N_i$;
- 6: *Iteration updates*
Calculate $q_{i,k}(l+1)$ via (2);
Calculate matrix $\Omega_{i,k}(l+1)$ via (2);
- 7: *Estimation updates*
Return estimate $\hat{x}_{i,k} = P_{i,k} q_{i,k}(L)$;
Return covariance $P_{i,k} = (\Omega_{i,k}(L))^{-1}$.

the number of neighbors. In addition, the induced weighted matrix of this graph is denoted as $\mathcal{A} = [\alpha_{ij}]_{N \times N} = \mathcal{S} + \text{diag}_{i,N} \{ \sum_{j=1}^N \alpha_{ij} \}$. Additionally, the shorthand $\text{diag}_{i,n} \{ M_i \}$ used in this paper denotes a block diagonal matrix with diagonal blocks being matrices $M_1, M_2, \dots, M_n$.

With the help of adopted consensus mechanisms and filter structures, Kalman-based distributed filtering algorithms can be roughly divided into the following six cases.

1) **Algorithm 1. Kalman-Consensus Filtering:** Inspired by the result in [11], a two-stage Kalman-consensus filtering algorithm is developed in [12]. This kind of algorithm described by (2) is composed of a classical local Kalman filter (1) and a consensus update of information matrices and vectors

$$\begin{cases} \Omega_{i,k}(l+1) = \sum_{j \in N_i} \pi_{ij} \Omega_{j,k}(l) \\ q_{i,k}(l+1) = \sum_{j \in N_i} \pi_{ij} q_{j,k}(l) \\ P_{i,k} = (\Omega_{i,k}(L))^{-1}, \quad \hat{x}_{i,k} = P_{i,k} q_{i,k}(L) \end{cases} \quad (2)$$

with $\Omega_{i,k}(0) = (P_{i,k}^- - K_{i,k} C_{i,k} P_{i,k}^-)^{-1}$ and $q_{i,k}(0) = \Omega_{i,k}(0) x_{i,k}^o$, where π_{ij} is the consensus weight and L is the selected step length of consensus calculation. Its pseudocode is provided in Algorithm 1.

Note that, even if the local sensor was undetectable, a sufficient condition is proposed in [13] to expose the uniform boundedness under the assumption of weightedly uniform detectability, which includes uniform detectability as a special case. Such an algorithm is further extended to two more general cases: the one is under the unscented Kalman filtering framework; and the other is under the cubature Kalman filtering (CKF) framework. For the first case, the boundedness condition of a consensus-based algorithm is derived in [14] with the help of a series of assumptions on system parameters. For the second one, an improved version, named as variational Bayesian consensus CKF, is proposed in [15], where the variational Bayesian approximation is utilized to estimate the covariance of measurement noises. Furthermore, a hybrid Kalman algorithm via CKF is designed in [16] by blending the consensus strategies on both measurements and information matrices. Finally, when the random link failure

Algorithm 2

- 1: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$, $P_{i,k}^-$ and $K_{i,k}$ via (1);
- 2: *Diffusion calculation*
- 3: *Communications*
Broadcast $y_{i,k}$ to all neighbors;
Receive $y_{j,k}$ from all neighbors $j \in N_i$;
- 4: *Calculations*
Calculate matrices $\Omega_{i,k}$ and $P_{i,k}$ via (3);
Calculate vectors $q_{i,k}$ and $\hat{\varphi}_{i,k}$ via (3);
- 5: *Communications*
Broadcast $\hat{\varphi}_{i,k}$ to all neighbors;
Receive $\hat{\varphi}_{j,k}$ from all neighbors $j \in N_i$;
- 6: *Estimation updates*
Return $\hat{x}_{i,k} = \sum_{j \in N_i} \pi_{ij} \hat{\varphi}_{j,k}$.

is a concern, in [17], a renewed weighted matrix is introduced in consensus steps to implement distributed filtering and the effect on boundedness from the statistics information of random link failures is thoroughly investigated.

2) **Algorithm 2. Diffusion Kalman Filtering:** The diffusion Kalman filtering is developed in [18] by replacing consensus strategies [12] with diffusion strategies [19] after the measurement update of the Kalman filter. The purpose of diffusion step $\hat{x}_{i,k} = \sum_{j \in N_i} \pi_{ij} \hat{\varphi}_{j,k}$ (see its information version (3) and Algorithm 2 for the corresponding pseudocode) is to approximate the global filtering performance via local node interactions

$$\begin{cases} \Omega_{i,k} = \sum_{j \in N_i} C_{i,k}^T R_j^{-1} C_{j,k}, \quad P_{i,k}^{-1} = (P_{i,k}^-)^{-1} + \Omega_{i,k} \\ \hat{x}_{i,k} = \sum_{j \in N_i} \pi_{ij} \hat{\varphi}_{j,k}, \quad q_{i,k} = \sum_{j \in N_i} C_{i,k}^T R_j^{-1} y_{j,k} \\ \hat{\varphi}_{i,k} = \hat{x}_{i,k}^- + P_{i,k} (q_{i,k} - \Omega_{i,k} \hat{x}_{i,k}^-). \end{cases} \quad (3)$$

In comparison with Algorithm 1, this strategy is independent of the consensus iteration between two consecutive Kalman filter updates and thus improves the fusion efficiency of new measurement information [20]. Based on this seminal research, diffusion Kalman filtering receives great attention [21] and the diffusion weights π_{ij} are further optimized through the well-known covariance intersection approach. Especially, when the uniform observability condition is satisfied under global measurements, an improved algorithm developed in [20] can guarantee the boundedness of estimation error covariances of each sensor under a certain connectivity condition of time-varying topologies.

3) **Algorithm 3. Kalman-Based Filtering I:** Based on the consensus-based Kalman filter designed in [11], a version relying on neighbors' estimate is proposed in [22] and the corresponding filter structure is constructed as follows:

$$\begin{cases} q_{i,k} = \sum_{j \in N_i} (\hat{x}_{j,k-1} - \hat{x}_{i,k-1}) \\ \hat{x}_{i,k} = \hat{x}_{i,k}^- + F_{i,k} A_k q_{i,k} + K_{i,k} (y_{i,k} - C_{i,k} \hat{x}_{i,k}^-) \end{cases} \quad (4)$$

Algorithm 3

-
- 1: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$ via (1);
 - 2: *Consensus error calculation*
 - 3: *Communications*
Broadcast $\hat{x}_{i,k-1}$ and $P_{ij,k}^-$ to all neighbors;
Receive $\hat{x}_{j,k-1}$ and $P_{ji,k}^-$ from all neighbors;
 - 4: *Gain calculations via similar formulas in [22]*
Select $F_{i,k}$ via a similar formula (18);
Calculate $K_{i,k}$ and $P_{ij,k}$ via a similar formula (13);
Calculate $P_{ij,k+1}^- = A_k P_{ij,k} A_k^T + Q$;
 - 5: *Error calculations*
Calculate $q_{i,k}$ via (4) in this paper;
 - 6: *Estimation updates*
Return $\hat{x}_{i,k}$ via (4).
-

where $q_{i,k}$ describes the consensus error of estimated states and the consensus gain $F_{i,k}$ is exploited to guarantee the stability. Additionally, the formulas on $K_{i,k}$, $P_{i,k}$, and cross-covariance $P_{ij,k}$ can be obtained by using the same method in [22] and hence omitted in this survey paper for the convenience of presentation. For the convenience of application, the pseudocode is provided in Algorithm 3. Obviously, the accumulative error of neighbors' estimate is utilized to achieve the consensus by replacing raw measurements and covariance information. In comparison with the local Kalman filter, the introduced accumulative error leads to the coupling of filtering error dynamics, and thereby the cross-covariance needs to be handled when discussing the stability of developed filtering algorithm. Nowadays, a suboptimal algorithm ignoring cross-covariance is developed in [23] to deal with the filtering issue with event-triggered communication protocols.

4) *Algorithm 4. Kalman-Based Filtering II:* Combing with the advantage of consensus schemes in Algorithms 2 and 3, a filtering algorithm is designed in [24] only via the weighted sum of neighbors' innovation

$$\begin{cases} q_{i,k} = \sum_{j \in N_i} \alpha_{ij} (y_{j,k} - C_{j,k} \hat{x}_{i,k}^-) \\ \hat{x}_{i,k} = \hat{x}_{i,k}^- + K_{i,k}^1 q_{i,k} + K_{i,k}^2 (y_{i,k} - C_{i,k} \hat{x}_{i,k}^-) \end{cases} \quad (5)$$

where matrices $K_{i,k}^1$ and $K_{i,k}^2$ are two gains and the corresponding pseudocode can be found in Algorithm 4.

Specially, the distributed recursive filtering issue is investigated in [24] for systems in the presence of time delays and uniform quantization as well as deception attacks. By resorting to the gradient-based method, the desired gains $K_{i,k}^1$ and $K_{i,k}^2$ are analytically designed so as to minimize the trace of an upper bound of filtering error covariance (FEC). Additionally, in light of the mathematical induction combined with eigenvalue analysis, the challenge from double gains is skilfully overcome in filtering performance analysis and a sufficient condition is derived to ensure the asymptotic boundedness of a sequence of error covariances. What particularly worth mentioning is that

Algorithm 4

-
- 1: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$ via (1);
 - 2: *Consensus error calculation*
 - 3: *Communications*
Broadcast $y_{i,k}$ and $C_{i,k}$ to all neighbors;
Receive $y_{j,k}$ and $C_{j,k}$ from all neighbors;
 - 4: *Gain calculations via similar formulas in [24]*
Calculate $P_{i,k}^-$ via a similar formula (11);
Calculate $K_{i,k}^1$ and $K_{i,k}^2$ via similar formula (12);
 - 5: *Error calculations*
Calculate $q_{i,k}$ via (5);
 - 6: *Estimation updates*
Return $\hat{x}_{i,k}$ via (5).
-

Algorithm 5

-
- 1: Initialize the weighted matrix $\mathcal{A}$ for all nodes;
 - 2: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$ via (1);
 - 3: *Calculations*
 - 4: *Communications*
Broadcast $y_{i,k}$ to all neighbors;
Receive $y_{j,k}$ from all neighbors $j \in N_i$;
 - 5: *Gain calculations via similar formulas in [26] (needing a computation center);*
Calculate upper bounds of $P_{i,k}^-$ and $P_{i,k}$ for all nodes via similar formulas (14a) and (14b);
Calculate gains K_k^{ij} via a similar formula (18);
 - 6: *Communications*
Receive K_k^{ij} from the computation center;
 - 7: *Estimation updates*
Return $\hat{x}_{i,k}$ via (6).
-

this algorithm possesses the high scalability with both low calculation and communication burden by sacrificing the consensus performance.

5) *Algorithm 5. Kalman-Based Filtering III:* In the above algorithm, all innovations are summed together with given weights and then employed to update the prediction. Such an algorithm cannot adequately identify the effect on filtering performance from different neighbors' information, especially, for the case with heterogeneous sensors. As such, a more general structure is designed in [25], and removing gain uncertainties leads to the following simplified version:

$$\hat{x}_{i,k} = \hat{x}_{i,k}^- + \sum_{j \in N_i \cup \{i\}} \alpha_{ij} K_k^{ij} (y_{j,k} - C_{i,k} \hat{x}_{i,k}^-) \quad (6)$$

where the formula K_k^{ij} , dependent on global error covariance $P_{i,k}$, can be derived along the same line in [25] and [26] and hence omitted in this survey paper for the simplicity of presentation. Obviously, in comparison with Algorithms 3 and 4, the gain K_k^{ij} provides more design freedom although the calculation burden is increased. The pseudocode is provided in Algorithm 5.

Algorithm 6

-
- 1: *Prediction updates*
Calculate $\hat{x}_{i,k}^-$ and $\hat{y}_{i,k}^-$ via (7);
 - 2: *Consensus error calculation*
 - 3: *Communications*
Broadcast $\hat{y}_{i,k}^-$ to all neighbors;
Receive $\hat{y}_{i,k}^-$ from the computation center;
 - 4: *Gain calculations via Algorithm 2 in [27]*
Calculate matrices G_k , $\tilde{A}_k$, $\hat{A}_k$, $\tilde{C}_{i,k}$ and $\hat{C}_{i,k}$;
Calculate $K_{i,k}$ and L_k^{ij} via Algorithm 2 in [27];
 - 5: *Error calculations*
Calculate $q_{i,k}$ via (7);
 - 6: *Estimation updates*
Make the pseudo-observation $z_{i,k}$ via (7);
Return $\hat{x}_{i,k}$ and $\hat{y}_{i,k}$ via (7).
-

6) Algorithm 6. “Consensus+Innovation” Filtering: By resorting to the pseudo-observation, a distributed estimator of pseudostate $G_k x_k$, a linear transformation of target states, is embedded into the fusion process based on transformed measurements, which results in the following algorithm:

$$\begin{cases} z_{i,k} = C_{i,k}^T R_i^{-1} y_{i,k}, & q_{i,k} = \sum_{j \in N_i} L_k^{ij} (\hat{y}_{j,k}^- - \hat{y}_{i,k}^-) \\ \hat{y}_{i,k} = \hat{y}_{i,k}^- + q_{i,k} + L_k^{ii} (z_{i,k} - (\tilde{C}_{i,k} \hat{y}_{i,k}^- - \hat{C}_{i,k} \hat{x}_{i,k}^-)) \\ \hat{x}_{i,k} = \hat{x}_{i,k}^- + K_{i,k} (\hat{y}_{i,k} - G_k \hat{x}_{i,k}^-) \\ \hat{y}_{i,k+1}^- = \tilde{A}_{i,k} \hat{y}_{i,k} + \hat{A}_{i,k} \hat{x}_{i,k}, & \hat{x}_{i,k+1}^- = A_{i,k} \hat{x}_{i,k} \end{cases} \quad (7)$$

where $q_{i,k}$ describes the weighted consensus error of predicted pseudostates. In above algorithm, some critical matrices are defined as $G_k = \sum_{i=1}^N C_{i,k}^T R_i^{-1} C_{i,k}$, $\tilde{A}_k = G_k A_k G_k^\dagger$, $\hat{A}_k = G_k A_k (I - G_k^\dagger G_k)$, $\tilde{C}_{i,k} = C_{i,k}^T R_i^{-1} C_{i,k} G_k^\dagger$, and $\hat{C}_{i,k} = C_{i,k}^T R_i^{-1} C_{i,k} (I - G_k^\dagger G_k)$, where $\dagger$ stands for the Moore–Penrose pseudoinverse. The formulas L_k^{ij} and $K_{i,k}$ can be found in [27], and the corresponding pseudocode is provided in Algorithm 6. In comparison with the distributed information Kalman filter developed in [28], the algorithm for the time-invariant case can converge to a bound only requiring global detectability and network connectivity.

Generally speaking, the calculation complexity and communication burden are two important indexes for above-discussed filtering algorithms. For Kalman-based algorithms, the calculation complexity at each instance is mainly dependent on the number of consensus steps, the inversion operation with $O(n^3)$ for an $n \times n$ square matrix, and the multiplication operation with $O(nsm)$ for an $n \times s$ matrix multiplying an $s \times m$ matrix. Meanwhile, the communication burden relies on the number of transmitted data. Assuming that $x_k \in \mathcal{R}^n$, one needs to implement 3 times of the matrix inversion operation, 12 times of the matrix multiplication operation, and the transmission of $\Theta_i n(n+2)(L+1)$ data for Algorithm 1, and $2\Theta_i + 2$, $4\Theta_i + 6$, and $2\Theta_i n$ for Algorithm 2. As such, Algorithm 1 possesses a low calculation complexity and Algorithm 2 has a low

TABLE I
PERFORMANCE OF KALMAN-BASED ALGORITHMS

	Algorithm	High	Middle	Low
Calculation burden	I			✓
	II		✓	
	III	✓		
	IV		✓	
	V	✓		
	VI	✓		
Communication burden	I		✓	
	II		✓	
	III	✓		
	IV			✓
	V	✓		
	VI			✓
Scalability	I	✓		
	II	✓		
	III		✓	
	IV	✓		
	V			✓
	VI		✓	

communication burden. In summary, by analyzing the iterative process, one can get that:

- 1) Algorithm 1 needs to carry out the L -step consensus calculation at each instant, Algorithm 3 depends on the cross-covariance of filtering errors, and Algorithm 4 has to execute complex calculation to obtain the filter gains;
- 2) Algorithm 3 and Algorithm 4 need to exchange cross-covariances and global error covariances, respectively; and
- 3) Algorithm 5 is highly dependent on global error covariances, which reduces its scalability (see Table I for more details).

Finally, recalling Algorithms 1 and 2, the selection of weight π_{ij} shows a considerable impact on filtering performance, and thereby the optimized weight design receives the persistent investigation (see [13], [14], and the references therein).

In the past few years, Algorithm 1 with $L = 1$ is employed in [29] to implement the dynamic state estimation for real-time monitoring of power systems. It should be pointed out that distributed estimators using local data effectively overcome the challenges from both communication latency and communication burden existing in a central case. Under the variational inference framework, a simplified version of Algorithm 2 is developed in [30] for distributed hybrid power state estimation, where an auto-encoder technique is adopted to reduce the data dimensionality in mixed measurements subject to phase errors. Algorithm 3 is employed to estimate the operating condition of renewable microgrids with reliable channels in [31] and packet losses in [32], and the conditions of corresponding convergence related on the Laplacian matrix are also analyzed simultaneously. For large-scale power networks, the merit of Algorithm 4 is clearly shown in [33] and the estimated state is utilized to compensate the information loss in distributed control due to the multirate nature of systems. Finally, according to a weighted least squares cost, a “consensus+innovation” based algorithm is proposed in [34] to estimate the vector of bus angles and converges almost surely to the centralized least squares estimator when the assumption of connectivity is satisfied.

TABLE II
DISTRIBUTED FILTER STRUCTURES

	Filter structure
I	$\hat{x}_{i,k+1} = A_k \hat{x}_{i,k} + \sum_{j \in N_i \cup \{i\}} \alpha_{ij} K_{ij,k} \gamma_{j,k}$
II	$\hat{x}_{i,k+1} = A_k \hat{x}_{i,k} + \sum_{j \in N_i \cup \{i\}} \alpha_{ij} (K_{i,k} \gamma_{j,k} + L_{i,k} \delta_{ij,k})$
III	$\hat{x}_{i,k+1} = \sum_{j \in N_i \cup \{i\}} \alpha_{ij} (K_{ij,k} \hat{x}_{j,k} + L_{ij,k} y_{j,k})$
IV	$\hat{x}_{i,k+1} = A_k \hat{x}_{i,k} + K_{i,k} \gamma_{i,k} + L_{i,k} \sum_{j \in N_i} \delta_{ij,k}$

B. Non-Kalman-Based Distributed Filtering

In parallel with the evolution of Kalman-based filtering algorithms, the finite-horizon distributed filtering of various time-varying systems receives an ever-increasing interest. Different from the consensus or diffusion strategies adopted in Kalman-based filtering algorithms, the improvement of filtering performance mainly relies on the distributed fusion of neighbors's information, which is realized via designed filter structures where the filter gains act as the role of fusion weights. Denote the innovation and the consensus errors as $\gamma_{i,k} = y_{i,k} - C \hat{x}_{i,k}$ and $\delta_{ij,k} = \hat{x}_{j,k} - \hat{x}_{i,k}$ and gains of the i th sensor as $K_{i,k}$, $L_{i,k}$, $K_{ij,k}$, and $L_{ij,k}$ for different scenarios. Taking the available information into account, some typically distributed filter structures are summarized in Table II. Obviously, Structure I is a typical Luenberger observer, and Structures II and IV can be regarded as its variant via replacing neighboring innovations by consensus errors. Furthermore, Structure III with most design parameters is usually employed to deal with the filtering issue of complex plants with unknown time delays and uncertainties as well as weak nonlinearities.

From a technical point of view, there are three representative approaches to design desired gains, namely the backward recursive Riccati difference equation (BRDE), the recursive linear matrix inequality (LMI), and the set-membership approach. With the help of the Moore–Penrose pseudoinverse, the BRDE approach is first developed in [35] by blending the finite-horizon H_∞ performance and the nominal H_2 cost, and further adopted to deal with some fashionable topics with various communication protocols (see [36] and the references therein). Furthermore, the recursive LMI tool is employed to handle the distributed filtering issue in [37], under which the conception of finite-horizon H_∞ -consensus is creatively defined to quantify bounded consensus regarding to filtering errors. In addition to these two approaches, the set-membership approach serves as another effective tool for handling filtering issues of time-varying systems. Different from popular point-estimation ones, such an approach provides a reliable interval involving true values and therefore satisfies various hard constraints from practical engineering. Here, we refer the readers to [38]–[40] for some most cutting-edge researches on this topic. It is worth pointing out that, in almost all literature considering randomly occurring network-induced phenomena, the augmentation of filtering errors is usually inevitable, which leads to the obtained design condition dependent on the global topology information. This typical characteristic extremely limits the application in large-scale sensor networks. In other words, in order to execute the filtering task online, the network scale is commonly required to

TABLE III
PERFORMANCE OF NON-KALMAN-BASED ALGORITHMS

	Structures	High	Middle	Low
Communication burden	I			✓
	II		✓	
	III		✓	
	IV		✓	
Scalability	I			✓
	II			✓
	III			✓
	IV			✓

be small and the topology information is usually open. Additionally, their calculation complexity is dependent on the number of dimensions of linear matrix inequalities and the number of decision variables. As such, Algorithm III owns the highest calculation complexity. Furthermore, Algorithms II–IV all need to transmit $2(\Theta_i + 1)n$ data and thus have the same communication burden. The concrete performance is summarized in Table III.

When the research concern is extended into infinite horizon, various distributed filtering schemes based on stability or H_∞ performance analysis were developed in recent years (see [41], [42], and the references therein). For instance, the distributed H_∞ filtering of polynomial nonlinear stochastic systems is investigated in [41] by constructing a polynomial Lyapunov function candidate, and the filter gains are designed in terms of the solution of parameter-dependent LMIs. For sensor networks with a randomly varying topology described by Markovian, some sufficient conditions are derived in [42] to satisfy the predetermined H_∞ level evaluating relative disagreement of estimated states both in the mean-square sense and with probability 1. Furthermore, some interesting results can be found in [43] for the case with event-triggered communication, in [44] for the case with sampled-data, and in [45] for the case subject to time delays. Compared with the finite-horizon case, the filter parameters can be obtained offline and therefore their scalability can be improved to some extent. Note that, due to the unavoidable impact from various network-induced phenomena, the obtained design condition applying to networked systems also relies on the augmented filtering error dynamics and their disadvantages in performance analysis are similar to that in finite-horizon cases. Additionally, the conservatism from time delays can be reduced possibly via various tools of time-delayed analysis adopted in [46]. Generally speaking, the application of these tools may sacrifice the computing performance partly.

Finally, let us briefly review the state-of-the-art of distributed filtering in the frameworks of Bayesian filtering and particle filtering (PF). Under the probability perspective, it is interesting to utilize the well-known Bayesian interpretation to disclose: what kind of posterior probability distribution of unknown states is reasonable in the formalization of distributed Kalman filtering; and how it is compared to the optimal posterior distribution of states conditioned on all present and past network data [47]. Usually, the locally available measurements $Y_{i,k} = \{y_{j,k}, j \in N_i \cup \{i\}\}$ together with mutual independence assumptions are first used to update the desired posterior PDF $\tilde{p}_i(x_k | Y_{i,k})$, and then the obtained PDFs are assimilated into

a PDF $p_i(x_k|Y_{i,k}) \propto \prod_{j \in N_i \cup \{i\}} [\tilde{p}_j(x_k|Y_{j,k})]^{\alpha_j}$ [48] via optimizing average Kullback-Leibler divergence [49] or Logarithmic opinion [50] where $\sum \alpha_j = 1$. Recently, various alternative schemes are developed to overcome the deficiency of the above-mentioned fusion approach in communication and computation. For instance, in the framework of consensus, alternating direction methods of multipliers combined with variational Bayesian algorithms are proposed to carry out the fusion in [51] and [52] for systems with non-Gaussian noises or Gaussian noises with unknown covariance. Additionally, discussing the Cramér–Rao lower bound can realize the efficient sensor selection [53], and adopting the structure of information filter can obtain a compact form of filtering algorithm [52].

It should be pointed out that the calculation of PDFs results in a great challenge of the implementation of distributed Bayesian filters. As an alternative scheme, their approximation can be performed via Monte Carlo simulation and importance sampling, which is the main conception of PF. In the distributed fashion, each sensor runs a localized PF, in existing many literatures, and then disseminates and fuses local information by resorting to various distributed schemes, such as consensus and gossip as well as diffusion [54]. Recently, the consensus-based algorithms are receiving considerable research attention due to no requirement of both routing protocols and global knowledge. According to the types of quantities, the developed algorithms can be roughly divided into three categories [55]: algorithms focusing on calculation of particle weights [54], algorithms focusing on calculation of posterior parameters [56], and those focusing on calculation of likelihood parameters [57]. Note that distributed PFs usually outperform standard Kalman filtering when the underlying plant and/or measurements models are nonlinear and/or non-Gaussian [58].

C. Power Schedule Control of Sensor Networks

In parallel with distributed filtering, significant efforts are devoted to the energy consumption minimization issues of sensor networks with the purpose of extending the lifespan via media access control protocols or sensor scheduling. Sensor scheduling in general refers to reducing the communication rate or energy consumption intentionally in order to achieve a desirable tradeoff between the filtering quality and the limited resource utilization [59], [60]. In other words, a suitable scheduling strategy of sensors should be adopted to effectively decrease the negative impact that forward error correction (FEC) is usually enlarged along with the reduced communication energy or communication rate. In the framework of Kalman filtering, various strategies are developed recently and can be roughly categorized as follows:

- 1) the FEC optimization under given scheduling constraints [61], [62];
- 2) the FEC optimization under given constraints on average energy usage [63], [64];
- 3) the optimization of energy usage under given FEC constraints [65]; and
- 4) the combination optimization between FEC and energy usage [66], [67].

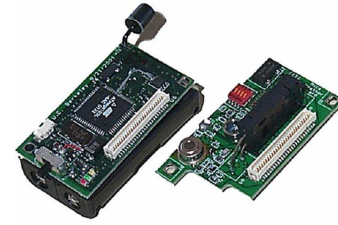


Fig. 2. Mica node.

For the purpose of analysis, a set of determinate or random binary variables is usually employed to indicate scheduled sensors, and the evolution of FEC is highly dependent on these binary variables. For issues with predetermined scheduling limitation, the sum of binary variables is subject to a given constraint describing the number of authorized sensors [62] or sensors with the high transmission power [61]. For finite-horizon scheduling, the characteristic of optimal policy is to schedule all transmission in a certain part of the horizon [61] or the limited number of transmission as uniformly as possible [68]. For an infinite horizon case, a metric of averaged FEC is generally adopted and some interesting algorithms, such as greedy algorithms and convex optimization approaches [62], [69], are developed to find optimal policies themselves or corresponding parameters therein (e.g., the ideal scheduling probability) in the limit case. Furthermore, the optimal policy can be approximated closely by a periodic scheduling with finite length [70]. On the other hand, considering the constraint on energy usage, a Markov decision process is utilized to reveal the structure of optimal scheduling policies and the derived sufficient and necessary condition of filtering stability depends on the high-energy packet reception ratio in [63] or the channel recovery rate in [64]. Unfortunately, it is indispensable for the feedback information from the remote estimator.

D. Hardware Platform

Sensor network platforms are likewise undergoing a revolution along with improving processors and memories to enhance the capabilities of local processing and computing, and various platforms with different CPUs and operating systems are designed to satisfy the commercial application (see [71] for more details). For example, as one of the most popular microsensor platforms, the Mica platform, which is designed by UC Berkeley, is with an Atmel ATmega processor running the TinyOS operating system (see Fig. 2 for its node [72]). In [73], three typical platforms of sensor networks, i.e., Particle, μ Node and Sindrion, are systematically designed by using a service-oriented approach and the first two ones are confirmed by a BP petrochemical plant in the U.K., where PIC18f6720 and MSP430 microcontrollers are integrated into sensor nodes (see Fig. 3 [73] for more details). In [74], a distributed Kalman filtering algorithm is implemented in the Tmote Sky sensor network together with a small ultrasound receiver in order to realize the localization, where the desired weights are optimized offline to yield a small estimation error covariance in stationarity. Recently, various hardware used for sensor networks and robot

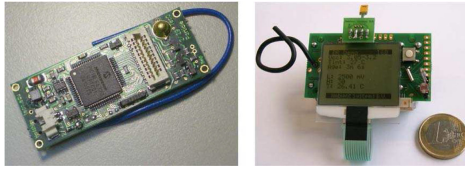


Fig. 3. Sensor node: Particle and μ Node.

platforms are thoroughly surveyed and the scope of services with different facilities are also discussed in [75]. Finally, the sequential discrete KF algorithm is performed in a field programmable gate array (FPGA) to obtain the real-time state estimate of active distribution networks [76].

III. DISTRIBUTED CONTROL FOR INDUSTRIAL CPSs

Distributed control is one of attractive technologies in industrial CPSs and widely implemented in various industrial manipulator systems as well as power grids.

A. Distributed Cooperative Control of Manipulators

Benefiting from higher degrees of freedom, the cooperation control of mobile manipulator systems is considered as an important technique to enhance efficiency and flexibility of industrial automation systems [77]. The kinematics and dynamics of manipulators can be modeled by an Euler–Lagrange (EL) equation [78] providing an ideal foundation of performance analysis and synthesis. Therefore, it becomes one of the emerging research topics in industrial CPSs, and some preliminary results can be found in the literature, see [79]–[82] to name a few. For instance, in [82], the distributed optimal tracking control with omnidirectional vision sensors is transformed into an optimal regulation issue of an integrated system, and an algorithm is developed accordingly to approximate the solution of corresponding Hamilton–Jacobi–Isaac equations.

1) *Distributed Control Based on CQP*: For redundant manipulators, the purpose of the inverse-kinematics issue, one of the most fundamental issues, is to generate an ideal trajectory in the joint space according to the desired trajectories of manipulator's end effector being provided in the Cartesian space [83]. Note that the number of equations in such a problem is less than the number of decision variables, which proposes extra design freedom and thus increases the applicability. In other words, this property offers more design selections with the concern of certain optimum criteria (e.g., minimum joint velocity $\|\dot{q}\|$ and tracking error $\|\dot{r}_d - \dot{q}\|$) and extra constraints from practical engineering requirements (e.g., joint limits and collision avoidance) [84]. In addition, solving the ideal trajectory can be easily reformulated as a constrained quadratic program (CQP) issue

$$\min_{\dot{q}} f(\dot{q}, \dot{r}_d - \dot{q}) \quad \text{s.t. } \dot{q} \in \Omega, \text{ and/or } g(J(q)) = 0$$

where f is a convex function and $g(J(q))$ stands for the Jacobian equality constraint. According to derived Karash–Kuhn–Tucker conditions, the optimal solution can be obtained via constructed recurrent neural networks (NNs) [85], [86]. When the local communication among neighbor manipulators is taken into consideration, the addressed issue has to be enslaved to a set of auxiliary

constraints only dependent on the neighbor information [83], [87], [88]. For instance, considering the limited communication bandwidth and communication range as well as electromagnetic interferences, a distributed scheme via a noise-tolerant zeroing NN is developed in [83] to solve a CQP problem constructed by a Lagrange multiplier method. Theoretical analysis reveals that the proposed scheme without noises can make position errors exponentially convergent.

2) *Distributed Control Based on NNs*: Different from solving a CQP issue, NNs or adaptive NNs can be utilized to approximate the nonlinear kinematics function of manipulators involving various complex cases, such as modeling uncertainties, input deadzones and output constraints as well as torque disturbances [89], [90]. The main idea is that an ideal controller is first derived via a constructed Lyapunov candidate and then its unknown part is approximated via NNs [91], [92]. For instance, $W\sigma(q, \dot{q}) = C(q, \dot{q})\dot{q} + G(q)$, where W is the NN weight, σ means the adopted activation function, $C(q, \dot{q})\dot{q}$ denotes the centripetal and Coriolis torques, and $G(q)$ stands for the gravitational force. For distributed cooperative control, the developed control scheme makes manipulators realize position and velocity synchronization while guaranteeing a formation. In addition, the designed controller has to embed an error term relying on both received information and communication topology to compensate the drift from predetermined ideal trajectory and, at the same time, a decomposition in the null space is employed to handle the challenges from nonintegrable and independent constraints in [85] and [93]. Under this conception, a vector can be artificially introduced to construct a dynamic equation of motion in reduced feasible-space [77], [85]. For instance, by resorting to a decoupled dynamics, the distributed control of both the end-effector motion and the mobile base motion is investigated in [77] to achieve different missions of mobile manipulators with a jointly connected topology. A master–slave framework based on NNs is adopted to handle the teleoperation subject to geometrically unknown constraints in [93] and both communication delays and input uncertainties in [85]. However, various network-induced phenomena and switching topologies as well as communication protocols have still not been fully taken into account mainly due to the difficulties of controller construction.

3) *Distributed Control Based on MASSs*: The cooperative control of multiple manipulators is adopted to realize the position and velocity synchronization while guaranteeing a formation or the joint trajectory for a primary task. In the past few years, multiple mobile manipulators are widely utilized in material transportation in modern factories, space exploration, and dangerous fields for dismantling bombs and moving nuclear infected objects [94]. Formation control of vehicles or distributed cooperative control of various industrial mobile manipulators can be transformed into the consensus issues of MASSs, each agent in which is described by an operational space model [77], [95]

$$\tilde{M}_i(q_i)\ddot{\theta}_i + \tilde{C}_i(q, \dot{q})\dot{\theta}_i + \tilde{G}(q_i) = \tilde{B}(q_i)\tau_i$$

where τ_i is the input torque vector and other parameter matrices are dependent on both the EL equations of the i th manipulators and the nonlinear mapping function between the joint

trajectory and the Cartesian coordinate. Recently, there are a rich body of results on consensus control of MASs with various networked-induced phenomena, switching network topologies, or sampling (see [96], [97], and the references therein). Furthermore, considering limited communication and energy resources, event-triggered consensus protocol receives considerable research attention and some interesting reviews on this topic are published in [98]. Specifically, the recent advancement is systematically surveyed in [99] according to different sampling mechanisms, especially event-triggered sampling with various thresholds. Based on the merit of event-triggered mechanism, the applications of consensus analysis in power sharing of microgrids and formation control of multirobot systems are summarized in [98]. It is worth noting that the fashionable results for linear multiagent systems cannot be effectively utilized to solve the specific challenges coming from the inherent nonlinearity of networked EL systems, not to mention that from various network-induced phenomena and communication protocols.

B. Distributed Model Predictive Control (DMPC) in Industrial CPSs

Benefiting from the merit of satisfaction of certain constraints, DMPC is receiving considerable research interest. Specifically, in the framework of DMPC, the transient performance can be effectively adjusted according to current and predicted information obtained by the solution of an optimization problem with some physical constraints. In this paper, the latest developments are roughly introduced according to the following three cases.

1) Case With Coupled Constraints: In many practical engineering, lots of distributed systems, such as power systems and cooperative robot systems, are governed by a group of uncoupled system dynamics but a possibly coupled constraint on global system states and control inputs [100]. This global constraint can be described by $\sum_{i=1}^m g_i(x_i, u_i) \leq 1$, where m is the number of subsystems (see [101] for more details). Recently, it is disclosed that the optimality of the overall system is dependent on a dual problem of the Lagrangian function. Under this dual problem, dual variables associated with the above-mentioned constraint can be regarded as consensus variables and therefore the solution of DMPC can be transformed into a distributed consensus optimization problem, which can be dealt with decomposition methods [102], alternating direction multiplier methods [103], dual accelerated gradient methods [104], and other optimization-based approaches. Among others, serial DMPC schemes [101], in which one subsystem is optimized at a time slot while holding all others constant. In other words, local optimization problems are solved sequentially in each subsystem, where neighboring interaction and/or solutions have to be taken into consideration. Unfortunately, the optimality applying this kind of methods is unclear although it offers greater flexibility in communication and computation.

DMPC with probabilistic coupled constraints also becomes a concern [105], [106], and the corresponding form is $\mathbb{P}\{\sum_{i=1}^m g_i(x^i, u^i) \leq b\} \geq p$, where b is a known threshold and p is the permitted probability. In contrast to deterministic forms, addressed issues allow constraint violations with the

occurring probability below a prespecified threshold. For this topic, the stochastic tube approach embedded in serial DMPC schemes is utilized to ensure the probabilistic constraint while satisfying the recursive feasibility and closed-loop stability (see [105] and the references therein).

2) Case With Coupled Dynamics: As is well known, the resultant feature of interconnection leads to significant difficulties for the application of model predictive control (MPC) in large-scale systems with coupled dynamics. The impact from neighboring states and/or control is generally handled by regarding them as a bounded mutual disturbance when performing the prediction of system behavior by resorting to both a nominal model of each subsystem and local interactions. Specifically, each subsystem transmits first its future reference trajectory to its neighbors while guaranteeing that the actual trajectory lies within a certain bound [107]. Some standard MPC tools are then employed to solve the addressed local optimization problem with various communication constraints. This route gives rise to some interesting tube DMPC approaches (see [108] for more details).

It is worth pointing out that most existing approaches can satisfy the primary concern on feasibility and stability, but expose two critical drawbacks: the conservatism induced by the boundedness of the mutual disturbance and the deficiency in global performance due to the local optimization. In order to overcome the conservatism drawback, some improved DMPC schemes are developed in recent years, such as DMPC with sharing contract sets [109], DMPC with terminal sets reconfigured online [110], and DMPC with switched cost functions [111]. At the same time, some prediction schemes are proposed to compensate the missing variables caused by event-triggered protocols and multirate mechanism in [112] and [113]. Furthermore, to improve the global performance, some iterative coordination-based algorithms are proposed to solve a centralized optimization problem in a distributed fashion [114].

3) Case With Coupled Cost Functions: The essential difference from above-mentioned two cases is that the connection among subsystems is realized by means of coupled cost functions [103], [115]

$$J_i = J(x_i, u_i, \{x_j, u_j\}), \quad j \in \mathcal{N}_j.$$

For instance, a dual-mode MPC strategy, which satisfies the feasibility and the stability for overall system, is developed in [116] to increase the robustness against the external disturbances. Note that the above result is dependent on a necessary assumption that a state-feedback control law of uncoupling subsystems is predetermined such that there is a constraint admissible invariant set, which plays an important role to guarantee the stabilization of closed-loop systems and the feasibility of developed DMPC schemes [115].

C. Distributed Control of Microgrids via Droop Characteristic

Microgrids, a kind of typical industrial CPSs, are usually an aggregation of distributed generators (such as renewable energy sources and conventional generators), energy storage systems,

and other equipment. Their upcoming, large-scale interconnection with the national primary grid requires urgent research attention to focus on highly reliable distributed regulation technologies. Especially, at the forefront of the field, a hierarchical control scheme, including the primary control and the secondary control, is conventionally adopted for this operation. The purpose of primary control, a decentralized control approach realized through a droop mechanism, is to stabilize microgrids and handle proportional load sharing among inverters. This control forces the bus voltages and frequency to deviate from their nominal values and could lead to poor reactive power sharing in the presence of unequal bus voltages. To overcome these shortcomings, secondary control under a distributed architecture is necessary to update the set points of the local primary control [117], [118].

1) Conventional Distributed Control Based on Droop Characteristics: Under the well-known d - q (direct quadrature) reference frame, the large-signal nonlinear dynamical model of microgrids with voltage source converters [119], [120] can be described in a compact form

$$\begin{cases} \dot{x}_i = f_i(x_i) + g_i(x_i)u_i + h_i(x_i)D_i \\ y_i^1 = v_{odi}, \quad y_i^2 = w_i = -d_i^p P_i + u_{2,i} \end{cases} \quad (8)$$

where y_i^1 and y_i^2 are outputs and x_i is the state vector usually selected as

$$x_i = [\theta_i \ P_i \ Q_i \ i_{Ldi} \ i_{Lqi} \ v_{odi} \ v_{oqi} \ i_{odi} \ i_{oqi}]^T$$

and other parameters can be found in [121]. Here, i_{Ldi} , i_{Lqi} , v_{odi} , v_{oqi} , i_{odi} , and i_{oqi} are the quadratic and direct components of converter current i_{Li} , LC filter voltage v_{di} , and output connector current i_{oi} . Furthermore, a more general 13th-order nonlinear model can be found in [122]. According to the partial feedback linearizing scheme, an auxiliary controller based on consensus errors constructed in [119] is

$$u_{i,\text{aux}} = -\kappa_i \left(\sum_{j \in N_i} \alpha_{ij} (v_{odj} - v_{odi}) + p_i (v_{odi} - v_{\text{ref}}) + \sum_{j \in N_i} \alpha_{ij} (d_j^q Q_j - d_i^q Q_i) \right) \quad (9)$$

for voltage control, where there exists at least one pinning gain $p_i > 0$ for voltage source converter i . In addition, taking proportional real power sharing into account, one has [123]

$$u_{i,\text{aux}} = -\kappa_i \left(\sum_{j \in N_i} \alpha_{ij} (\omega_j - \omega_i) + p_i (\omega_i - \omega_{\text{ref}}) + \sum_{j \in N_i} \alpha_{ij} (d_j^p P_j - d_i^p P_i) \right) \quad (10)$$

for frequency control. It should be pointed out that the regulating error of active power allocation $\sum_{j \in N_i} \alpha_{ij} (d_j^p P_j - d_i^p P_i)$ is embedded in the control strategy with intent to maintain the equal cost increment achieved in the primary control [123].

By utilizing iterative learning approaches, a robust distributed secondary control is presented in [124] to achieve both active power sharing accuracy and voltage/frequency restoration of microgrids with uncertain communication links. Furthermore, an improved controller by combining tracking performance and regulator synchronization is developed in [123] for islanded microgrids with a sparse communication network, and in [125] for inverter-based ac microgrids with additive channel noises. A consensus-based distributed controller is designed in [120] to provide more robustness to unmodeled dynamics, unknown disturbances, and uncertainties while restoring the desired voltage magnitude and frequency within finite time. Generally speaking, the controller gain usually depends on the eigenvalues of a Laplacian matrix, which are essentially global information. In contrast to these existing distributed methods, a fully distributed adaptive controller designed in [119] only utilizes the dynamics model of distributed generation (DG) units and neighbors' information and therefore eliminates the need of a central computing and communication unit. Furthermore, both single and multiple pinning schemes are discussed in [126] and it is disclosed that the selection of a pinning set depends on both the degree of algebraic connectivity and the distance of leaders. However, the problems on the selection of pinning nodes and the design of pinning strength are not solved effectively, especially for the cases with network-induced phenomena and communication protocols.

2) Distributed Control Based on Averaging Conception:

With the help of the standard frequency/active-power (f/P) and voltage/reactive-power (V/Q) properties, one has the following conventional controller structure with the conception of distributed averaging [127], [128] to adjust the inverter frequency ω_i and the amplitude E_i of output voltage:

$$\omega_i = \omega^* - d_i^p P_i - u_i^\omega, \quad E_i = E^* - d_i^q Q_i - u_i^E \quad (11)$$

where ω^* and E^* are the nominal network frequency and the nominal voltage of microsource i . A feasible distributed secondary control [128] is

$$\begin{cases} \kappa_i^\omega \dot{u}_i^\omega = -(\omega_i - \omega^*) - \sum_{j \in N_i} \alpha_{ij} (u_i^\omega - u_j^\omega) \\ \kappa_i^E \dot{u}_i^E = -\beta_i (E_i - E^*) - \sum_{j \in N_i} \alpha_{ij} \left(\frac{Q_i}{Q_i^*} - \frac{Q_j}{Q_j^*} \right) \end{cases} \quad (12)$$

where β_i , κ_i^ω , and κ_i^E are positive gains, Q_i^* is the reactive power rating of DG i , and α_{ij} is the ij th element of microgrid's Laplacian matrix.

Different from structure (10), the diffusive averaging term $\sum_{j \in N_i} \alpha_{ij} (u_i^\omega - u_j^\omega)$ is introduced into the above controller to maintain active power sharing. This kind of controllers is also named as distributed-averaging proportional-integral ones. An early version developed in [129] relies on both average voltage and average frequency of DG units and therefore there exists a forced assumption on the communication among all DGs. Fortunately, such a shortcoming is overcome via the above controller. As stated in [128], the designed distributed control removing the need of a central supervisory satisfies a tunable tradeoff

between reactive power sharing and voltage regulation. In addition, in light of the Lyapunov stability theory, some sufficient conditions are derived to evaluate the system stability and robustness to incomplete communication links.

3) Distributed Control Based on Angle Droop Characteristics: Similar to the above-mentioned droop control scheme, one has the following angle droop control model for a microgrid [130], [131] with purely inductive lines

$$d_i^p \dot{\theta}_i = \Delta P_i^* - \sum_{j=1}^N l_{ij} \sin(\theta_i - \theta_j) - u_i \quad (13)$$

where $d_i^p > 0$, θ_i , u_i , and l_{ij} denote, respectively, the droop coefficient, the bus phase angle, the control input determined by a secondary control loop, and the magnitude of pure imaginary admittance of the interconnecting line between inverters i and j . ΔP_i^* represents the inverter power mismatch between the nominal injection setpoint of inverters and the load demand of buses. To eliminate this angle deviation, a typical distributed secondary controller is

$$\kappa \dot{u}_i = d_i^p \dot{\theta}_i - \sum_{j=1}^N \alpha_{ij} \left(\frac{u_i}{d_i^p} - \frac{u_j}{d_j^p} \right)$$

where $\kappa > 0$ is the coefficient and α_{ij} is the same with one in (12). This distributed controller is employed in [132] to offer the best combination of flexibility and performance by using communication among generation units. Especially, when $\theta_i \simeq \theta_j$, the above control model can be simplified as (12).

Lately, stability and convergence properties receive considerable research efforts (see, e.g., [131] and [133]). Specially, a necessary and sufficient condition is derived in [131] to disclose the existence and uniqueness of locally exponentially stable equilibrium. In addition, frequency regulation without assuming time-scale separation can be ensured and ultimate boundedness can also be satisfied under a condition on power mismatch [130]. In particular, a simplified distributed controller is designed in [133] to steer the microgrid with security constraints to a desired steady state. The designed controller is also robust for loss of communication links and failures of distributed energy resources.

4) Distributed Control Based on State-Space Models: In addition, by regarding each control area as a single-machine single-load system or using the droop characteristics of power sharing, a state-space model of power systems can be described by

$$\dot{x}_i = A_{ii}x_i + \sum_{j=1, j \neq i}^N A_{ij}x_j + Bu_i + \Gamma_i n_{di} \quad (14)$$

where state vector x_i can be selected as $x_i = [\Delta\omega_i \Delta P_{ti} \Delta P_{gi} \Delta P_{tie}^i]^T$ for multiarea power systems [134], [135] augmenting the deviation of frequency ω_i , generator mechanical power P_{ti} , turbine valve position P_{gi} , and the net tie-line power flow ΔP_{tie}^i , or selected as $x_i = [V_i I_{ti} \phi_V \phi_I \varphi_V \varphi_I]^T$ for dc microgrids [33], [136], [137] reflecting the load voltage at point of common coupling, the filter current, the dynamics of primary control (the third and fourth states), and the dynamics of secondary voltage

controllers (the last two states). Additionally, n_{di} stands for a load disturbance and A_{ij} represents the coupled matrix of interconnected areas. The adopted distributed control is with the form [134], [135]

$$u_i = K_i x_i + \sum_{j=1, j \neq i}^N K_{ij} x_j$$

or the form based on consensus control [136]

$$u_i = K_i \sum_{j=1, j \neq i}^N \alpha_{ij} (\hat{x}_j - \hat{x}_i)$$

where $\hat{x}_i$ is the estimated state and K_i and K_{ij} are two controller gains to be designed. Furthermore, one can obtain a model similar to (14) by using small signal analysis under the d - q (direct quadrature) reference frame (see [138] and the references therein).

For large-scale power systems with limited bandwidth constraints, a delay-dependent stability criterion is proposed in [135] for systems subject to communication delays, and the relationship between delay margins and control gains is exposed in detail. Furthermore, the impact on the closed-loop system performance from both time delays and varying communication topologies is examined in [134] via multiagent technologies, and the dynamic performance under the event-triggered scheme is also analysed via simulations. In order to reduce the communication burden, event-triggered protocols dependent on dynamic consensus algorithms are designed in [136] and [137] to achieve both voltage regulation and proportional current (or load) sharing. With the help of constructed consensus estimators combined with the Lyapunov stability theory, the desired event-triggering conditions are obtained and the convergence and stability of the proposed dynamic consensus algorithm are discussed. In comparison with the condition in [136], the setup of the adaptive strategy in [137] results in a requirement of GPS signal.

IV. DISTRIBUTED SECURITY CONTROL AND FILTERING OF INDUSTRIAL CPSs

There is no doubt that the increased interaction between physical realms and cyber will result in unavoidable security vulnerabilities. In real-world CPSs, typical cyber-attacks include denial of service attacks, replay attacks, deception attacks, and so forth. Any successful attacks against industrial CPSs may cause the serious impact on the human society and national economy [139]. As such, the security of industrial CPSs to defend against these attacks is an urgent research concern [140]–[142]. It is worth mentioning that, compared with relatively mature techniques under traditional distributed frameworks, distributed control and filtering with security perspective still remain at an infant stage. Recently, from the perspective of defenders, the effect on state estimation from attacks was investigated (see [143], [144], and the references therein). Usually, the security can be enhanced by effectively utilizing the identification information of cyber-attacks or necessary attack information (i.e., attack intensity or attack frequency).

In light of the Kalman filtering theory with generalized cumulative sum algorithms, a distributed state estimator is designed in [145] in an attempt to realize a quick attack mitigation and system recovery. In comparison with the existing results based on least square approaches, the proposed scheme can not only successfully recognize bad data but also identify structured false data injection attacks. Besides, according to the well-known gradient descent, a recursive algorithm is designed in [24] to handle a distributed filtering issue over sensor networks subject to deception attacks and a sufficient condition is derived to check the stability of proposed algorithm. A distributed filter owning the capabilities of attack detection and state estimation is designed and tested via the wide-area monitoring of a power network in [146], where a hybrid Bernoulli random set is introduced to describe the joint information on the attack's presence or not. Different from distributed filtering algorithms in the mean square sense [147], the paradigm of Kullback–Leibler fusion is utilized to handle the challenge from probability density function. It should be pointed out that the obtained algorithm only has limited robustness to attacks arising from the assumption on attack intensity.

In the context of industrial CPS security, the performance indexes mainly include security and stability as well as resilience where the tradeoff between security and stability receives growing attention in the literature [9]. Similar to the distributed filtering of industrial CPSs discussed above, it is also urgent to mitigate the threat from various cyber-attacks via designed control strategies. In view of the core of critical infrastructures, the resulting successful attack on industrial CPSs is generally more serious in contrast to attacks on traditional networked control systems. In addition, according to system recovery, two schemes of attack-resilient cooperative control for industrial CPSs consisting of distributed agents are investigated in [148] via the identification and isolation of the misbehaving cyber elements and the cascade protection design of lossless systems. For operator-vehicle networks remotely maneuvered by an operator, a distributed resilient algorithm is developed in [149] to steer vehicles to the desired formation against attackers. The designed control algorithm can be effectively performed under a distributed form by resorting to the introduction of an auxiliary receding-horizon control. A cooperative resilient control strategy is introduced in [148] to regulate the active power from a cluster of DGs at a certain ratio of their maximal available power. Such a strategy combined with a proper observation networks can effectively monitor the behaviors of all neighbors and isolate the misbehaving DGs coming from noncolluding cyber-attacks. Additionally, a resilient distributed algorithm is proposed in [150] to solve a dc optimal power flow issue against data integrity attacks. In these two papers, the consensus conception is utilized to estimate the utilization ratios of DGs in [148] by introducing a virtual power plant and estimate the bus power imbalance in [150] by adopting the primal-dual decomposition method. It should be pointed out that almost all results on security in the framework of control theory are based on an assumption that system dynamics need to be simple, which leads to a gap between theoretical results and practical engineering applications.

V. CONCLUSION AND CHALLENGING ISSUES

A survey on distributed control and filtering has been provided for industrial CPSs described by differential dynamics models. For real-time monitoring, some typical Kalman-based distributed algorithms and various filter structures have been systematically summarized and their performances have been also discussed in detail. For distributed closed-loop systems, the development of distributed control strategies has been well addressed, especially for mobile manipulator systems and power systems with the requirements of power sharing and voltage and frequency regulation. However, there are still various limitations from communication links, bandwidth, computational burden and scalability as well as special engineering requirements, which potentially offer some scope for improving existing results or methodologies. Additionally, we pay more attention to these limitations and propose some important and yet challenging research topics, which sheds insightful light on the further research.

- 1) For practical CPSs, resource constraints or a practical function, such as communication bandwidth, limited energy, or plug-and-play, usually need to be taken into consideration. Under these constraints, it is no longer effective for the study of distributed control and filtering via established methods dependent on the eigenstructure of the coupling matrix. In other words, most of existing research results are based on some specific assumptions on topology structures and therefore there is an unavoidable stumbling block for practical applications. A feasible idea is to find a suitable condition of connectivity for randomly varying topologies or a condition on dwell time for general switching topologies to guarantee the stability of addressed systems.
- 2) When occurring above-mentioned resource constraints, existing distributed filtering schemes cannot simultaneously satisfy the requirements of computation burden, communication burden, and scalability so far. The main reason could come from the calculation of cross-covariances, the utilization of augmentation approaches, or consensus iteration. In addition, in light of its reliability in describing filtering accuracy, the filtering performance in probability sense is more suitable for practical engineering with various random phenomena. Unfortunately, it has been largely omitted probably due to the difficulty in mathematic analysis.
- 3) According to the characteristic of line impedance, Q/V and P/f droop control schemes are usually utilized in a microgrid with inductive lines, and Q/F and P/V ones are for the case with resistive lines. When considering a microgrid subject to complex line impedance or inaccurate P or Q regulation, the existing strategies for above two cases cannot guarantee a reliable control performance. The main reason could arise from the challenges of coupled P and Q equations. Hence, it is of significance to develop some novel distributed control strategies for more general power systems, which deserves deep investigation.

- 4) It is noteworthy that multiple pinning control has been regraded as an effective scheme in distributed secondary control of microgrids. However, to the best of the authors' knowledge, the problems on the selection of pinning nodes and the design of pinning strength are not solved effectively, especially for the cases with network-induced phenomena and communication protocols. Note that such a control issue presents the intermittent feature due to the data sparsity induced by above cases, and thereby it is a meaningful attempt to carry out the performance analysis in the framework of intermittent control.
- 5) Due to the vulnerability, cyber-attacks can be regarded as one of the major threats of CPSs. Up to date, some interesting results have been developed to detect whether the discussed system is subject to attacks. Unfortunately, the detected results cannot be effectively utilized to improve the performance of control or filtering. In other words, it is considerable to integrate the attack detection into the designed distributed filtering and control strategies against various cyber-attacks. In light of its good statistical characteristic, the χ^2 detector combined with an indicator function could be one of the best options.
- 6) Further research topics also include to deepen practical engineering applications of the developed theory, such as approximate dynamic control of mobile cooperative manipulator systems, distributed energy management of smart grids, supervisory control and data acquisition of power systems, and industrial process control of metallurgy and petrochemistry areas.

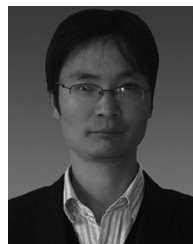
REFERENCES

- [1] Y. Turygin, P. Božek, Y. Nikitin, E. Sosnovich, and A. Abramov, "Enhancing the reliability of mobile robots control process via reverse validation," *Int. J. Adv. Robot. Syst.*, vol. 13, no. 6, pp. 1–8, Dec. 2016.
- [2] P. Božek, P. Bezák, Y. Nikitin, G. Fedorko, and M. Fabian, "Increasing the production system productivity using inertial navigation," *Manuf. Technol.*, vol. 15, no. 3, pp. 274–278, Jan. 2015.
- [3] A. W. Colombo, S. Karnouskos, O. Kaynak, Y. Shi, and S. Yin, "Industrial cyberphysical systems: A backbone of the fourth industrial revolution," *IEEE Ind. Electron. M.*, vol. 11, no. 1, pp. 6–16, Mar. 2017.
- [4] P. Leita, S. Karnouskos, L. Ribeiro, J. Lee, T. Strasser, and A. W. Colombo, "Smart agents in industrial cyber-physical systems," *Proc. IEEE*, vol. 104, no. 5, pp. 1086–1101, May 2016.
- [5] X. Feng, X. Kong, and Z. Xu, "Cyber-physical control over wireless sensor and actuator networks with packet loss," in *Wireless Networking Based Control*. New York, NY, USA: Springer, 2011, pp. 85–102.
- [6] X. Jiang, and S. Li, "Plume front tracking in unknown environments by estimation and control," *IEEE Trans. Ind. Inform.*, vol. 15, no. 2, pp. 911–921, Feb. 2019, doi: 10.1109/TII.2018.2831225.
- [7] D. Ding, Z. Wang, Q.-L. Han, and G. Wei, "Neural network based output-feedback control under Round-Robin scheduling protocols," *IEEE Trans. Cybern.*, to be published, doi: 10.1109/TCYB.2018.2827037.
- [8] C. Peng and Q.-L. Han, "On designing a novel self-triggered sampling scheme for networked control systems with data losses and communication delays," *IEEE Trans. Ind. Electron.*, vol. 63, no. 2, pp. 1239–1248, Feb. 2016.
- [9] D. Ding, Z. Wang, Q.-L. Han, and G. Wei, "Security control for a class of discrete-time stochastic nonlinear systems subject to deception attacks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 48, no. 5, pp. 779–789, May 2018.
- [10] C. Chen, J. Yan, N. Lu, Y. Wang, X. Yang, and X. Guan, "Ubiquitous monitoring for industrial cyber-physical systems over relay-assisted wireless sensor networks," *IEEE Trans. Emerg. Topics Comput.*, vol. 3, no. 3, pp. 352–362, Jul. 2015.
- [11] R. Olfati-Saber and J. S. Shamma, "Consensus filters for sensor networks and distributed sensor fusion," in *Proc. 44th IEEE Conf. Decis. Control*, Seville, Spain, Dec. 2005, pp. 6698–6703.
- [12] G. Battistelli, L. Chisci, G. Mugnai, A. Farina, and A. Graziano, "Consensus-based linear and nonlinear filtering," *IEEE Trans. Autom. Control*, vol. 60, no. 5, pp. 1410–1415, May 2015.
- [13] W. Li, G. Wei, D. W. C. Ho, and D. Ding, "A weightedly uniform detectability for sensor networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5790–5796, Nov. 2018.
- [14] W. Li, G. Wei, F. Han, and Y. Liu, "Weighted average consensus-based unscented Kalman filtering," *IEEE Trans. Cybern.*, vol. 46, no. 2, pp. 558–567, Feb. 2016.
- [15] K. Shen, Z. Jing, and P. Dong, "A consensus nonlinear filter with measurement uncertainty in distributed sensor networks," *IEEE Signal Process. Lett.*, vol. 24, no. 11, pp. 1631–1635, Nov. 2017.
- [16] Q. Chen, C. Yin, J. Zhou, Y. Wang, X. Wang, and C. Chen, "Hybrid consensus-based cubature Kalman filtering for distributed state estimation in sensor networks," *IEEE Sensors J.*, vol. 18, no. 11, pp. 4561–4569, Jun. 2018.
- [17] Q. Liu, Z. Wang, X. He, and D.-H. Zhou, "On Kalman-consensus filtering with random link failures over sensor networks," *IEEE Trans. Autom. Control*, vol. 63, no. 8, pp. 2701–2708, Aug. 2018.
- [18] F. S. Cattivelli, and A. H. Sayed, "Diffusion strategies for distributed Kalman filtering and smoothing," *IEEE Trans. Autom. Control*, vol. 55, no. 9, pp. 2069–2084, Sep. 2010.
- [19] C. G. Lopes and A. H. Sayed, "Diffusion least-mean squares over adaptive networks: Formulation and performance analysis," *IEEE Trans. Signal Process.*, vol. 56, no. 7, pp. 3122–3136, Jul. 2008.
- [20] J. Hu, L. Xie, and C. Zhang, "Diffusion Kalman filtering based on covariance intersection," *IEEE Trans. Signal Process.*, vol. 60, no. 2, pp. 891–902, Feb. 2012.
- [21] Y. Chen, G. Qi, Y. Li, and A. Sheng, "Diffusion Kalman filtering with multi-channel decoupled event-triggered strategy and its application to the optic-electric sensor network," *Inf. Fusion*, vol. 36, pp. 233–242, Jul. 2017.
- [22] R. Olfati-Saber, "Kalman-consensus filter: Optimality, stability, and performance," in *Proc. 48th IEEE Conf. Decis. Control, 28th Chin. Control Conf.*, Shanghai, China, Dec. 2009, pp. 7036–7042.
- [23] C. Zhang, and Y. Jia, "Distributed Kalman consensus filter with event-triggered communication: Formulation and stability analysis," *J. Franklin Inst.*, vol. 354, pp. 5486–5502, Sep. 2017.
- [24] D. Ding, Z. Wang, D. W. C. Ho, and G. Wei, "Distributed recursive filtering for stochastic systems under uniform quantizations and deception attacks through sensor networks," *Automatica*, vol. 78, pp. 231–240, Apr. 2017.
- [25] Q. Liu, Z. Wang, X. He, G. Ghinea, and F. E. Alsaadi, "A resilient approach to distributed filter design for time-varying systems under stochastic nonlinearities and sensor degradation," *IEEE Trans. Signal Process.*, vol. 65, no. 5, pp. 1300–1309, Mar. 2017.
- [26] Q. Liu, Z. Wang, X. He, and D.-H. Zhou, "Event-based recursive distributed filtering over wireless sensor networks," *IEEE Trans. Autom. Control*, vol. 60, no. 9, pp. 2470–2475, Sep. 2015.
- [27] S. Das, and J. M. F. Moura, "Consensus+innovations distributed Kalman filter with optimized gains," *IEEE Trans. Signal Process.*, vol. 65, no. 2, pp. 467–481, Jan. 2017.
- [28] S. Das, and J. M. F. Moura, "Distributed Kalman filtering with dynamic observations consensus," *IEEE Trans. Signal Process.*, vol. 63, no. 17, pp. 4458–4473, Sep. 2015.
- [29] M. Rostami and S. Lotfifard, "Distributed dynamic state estimation of power systems," *IEEE Trans. Ind. Inform.*, vol. 14, no. 8, pp. 3395–3404, Aug. 2018.
- [30] J. Du, S. Ma, Y.-C. Wu, and H. V. Poor, "Distributed hybrid power state estimation under PMU sampling phase errors," *IEEE Trans. Signal Process.*, vol. 62, no. 16, pp. 4052–4063, Aug. 2014.
- [31] M. Rana, L. Li, S. W. Su, and W. Xiang, "Microgrid state estimation: A distributed approach," *IEEE Trans. Ind. Inform.*, vol. 14, no. 8, pp. 3368–3375, Aug. 2018.
- [32] M. Rana, L. Li, and S. W. Su, "Distributed state estimation over unreliable communication networks with an application to smart grids," *IEEE Trans. Green Commun. Netw.*, vol. 1, no. 1, pp. 89–96, Mar. 2017.
- [33] M. S. Sadabadi, Q. Shafiee, and A. Karimi, "Plug-and-play robust voltage control of DC microgrids," *IEEE Trans. Smart Grid*, vol. 9, no. 16, pp. 6886–6896, Jun. 2018.

- [34] S. Kar, G. Hug, J. Mohammadi, and J. M. F. Moura, "Distributed state estimation and energy management in smart grids: A consensus innovations approach," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 6, pp. 1022–1038, Dec. 2014.
- [35] D. Ding, Z. Wang, H. Dong, and H. Shu, "Distributed H_∞ state estimation with stochastic parameters and nonlinearities through sensor networks: The finite-horizon case," *Automatica*, vol. 48, no. 8, pp. 1575–1585, Aug. 2012.
- [36] L. Sheng, Z. Wang, L. Zou, and F. E. Alsaadi, "Event-based H_∞ state estimation for time-varying stochastic dynamical networks with state- and disturbance-dependent noises," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2382–2394, Oct. 2017.
- [37] B. Shen, Z. Wang, and Y. S. Hung, "Distributed consensus H_∞ filtering in sensor networks with multiple missing measurements: The finite-horizon case," *Automatica*, vol. 46, no. 10, pp. 1682–1688, Oct. 2010.
- [38] X. Ge, Q.-L. Han, and Z. Wang, "A dynamic event-triggered transmission scheme for distributed set-membership estimation over wireless sensor networks," *IEEE Trans. Cybern.*, vol. 49, no. 1, pp. 171–183, Jan. 2019.
- [39] X. Ge, Q.-L. Han, and F. Yang, "Event-based set-membership leader-following consensus of networked multi-agent systems subject to limited communication resources and unknown-but-bounded noise," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5045–5054, Jun. 2017.
- [40] F. Yang, N. Xia, and Q.-L. Han, "Event-based networked islanding detection for distributed solar PV generation systems," *IEEE Trans. Ind. Inform.*, vol. 13, no. 1, pp. 322–329, Feb. 2017.
- [41] B. Shen, Z. Wang, Y. S. Hung, and G. Chesi, "Distributed H_∞ filtering for polynomial nonlinear stochastic systems in sensor networks," *IEEE Trans. Ind. Electron.*, vol. 58, no. 5, pp. 1971–1979, May 2011.
- [42] V. Ugrinovskii, "Distributed robust estimation over randomly switching networks using H_∞ consensus," *Automatica*, vol. 49, no. 1, pp. 160–168, Jan. 2013.
- [43] X. Ge, Q.-L. Han, and Z. Wang, "A threshold-parameter-dependent approach to designing distributed event-triggered H_∞ consensus filters over sensor networks," *IEEE Trans. Cybern.*, vol. 49, no. 4, pp. 1148–1159, Apr. 2019.
- [44] B. Shen, Z. Wang, and X. Liu, "A stochastic sampled-data approach to distributed H_∞ filtering in sensor networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 58, no. 9, pp. 2237–2246, Sep. 2011.
- [45] X.-M. Zhang and Q.-L. Han, "State estimation for static neural networks with time-varying delays based on an improved reciprocally convex inequality," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 1376–1381, Apr. 2018.
- [46] J. Wang, X.-M. Zhang, and Q.-L. Han, "Event-triggered generalized dissipativity filtering for neural networks with time-varying delays," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 77–88, Jan. 2016.
- [47] M. G. S. Bruno, and S. S. Dias, "A Bayesian interpretation of distributed diffusion filtering algorithms," *IEEE Signal Process. Mag.*, vol. 35, no. 3, pp. 118–123, May 2018.
- [48] K. Dedecius and P. M. Djurić, "Sequential estimation and diffusion of information over networks: A Bayesian approach with exponential family of distributions," *IEEE Trans. Signal Process.*, vol. 65, no. 7, pp. 1705–1809, Apr. 2017.
- [49] G. Battistelli and L. Chisci, "Kullback–Leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability," *Automatica*, vol. 50, no. 3, pp. 707–718, Mar. 2014.
- [50] S. Bandyopadhyay, and S.-J. Chung, "Distributed Bayesian filtering using logarithmic opinion pool for dynamic sensor networks," *Automatica*, vol. 97, pp. 7–17, Nov. 2018.
- [51] J. Hua, and C. Li, "Distributed outlier-robust Bayesian filtering for state estimation," *IEEE Trans. Signal Inf. Process. Netw.*, to be published, doi: 10.1109/TSIPN.2018.2889579.
- [52] P. Dong, Z. Jing, H. Leung, K. Shen, and M. Li, "Robust consensus nonlinear information filter for distributed sensor networks with measurement outliers," *IEEE Trans. Cybern.*, to be published, doi: 10.1109/TCYB.2018.2850368.
- [53] J. Li, F. Deng, and J. Chen, "A fast distributed variational Bayesian filtering for multisensor LTV system with non-gaussian noise," *IEEE Trans. Cybern.*, to be published, doi: 10.1109/TCYB.2018.2815697.
- [54] A. Mohammadi, and A. Asif, "Distributed particle filter implementation with intermittent/irregular consensus convergence," *IEEE Trans. Signal Process.*, vol. 61, no. 10, pp. 2572–2587, May 2013.
- [55] O. Hlinka, F. Hlawatsch, and P. M. Djurić, "Distributed particle filtering in agent networks," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 61–81, Jan. 2013.
- [56] D. Gu, "Distributed EM algorithm for Gaussian mixtures in sensor networks," *IEEE Trans. Neural Netw.*, vol. 19, no. 7, pp. 1154–1166, Jul. 2008.
- [57] O. Hlinka, F. Hlawatsch, and P. M. Djurić, "Consensus-based distributed particle filtering with distributed proposal adaptation," *IEEE Trans. Signal Process.*, vol. 62, no. 10, pp. 3029–3041, Jun. 2014.
- [58] S. S. Dias and M. G. S. Bruno, "Distributed Bernoulli filters for joint detection and tracking in sensor networks," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 2, no. 3, pp. 260–275, Sep. 2016.
- [59] Y. Shu, K. G. Shin, J. Chen, and Y. Sun, "Joint energy replenishment and operation scheduling in wireless rechargeable sensor networks," *IEEE Trans. Ind. Inform.*, vol. 13, no. 1, pp. 125–134, Feb. 2017.
- [60] J. Wu, X. Ren, D. Han, D. Shi, and L. Shi, "Finite-horizon Gaussianity-preserving event-based sensor scheduling in Kalman filter applications," *Automatica*, vol. 72, pp. 100–107, Oct. 2016.
- [61] L. Shi and L. Xie, "Optimal sensor power scheduling for state estimation of Gauss–Markov systems over a packet-dropping network," *IEEE Trans. Signal Process.*, vol. 60, no. 5, pp. 2701–2705, May 2012.
- [62] C. Li and N. Elia, "Stochastic sensor scheduling via distributed convex optimization," *Automatica*, vol. 58, pp. 173–182, Aug. 2015.
- [63] Z. Ren, P. Cheng, J. Chen, L. Shi, and H. Zhang, "Dynamic sensor transmission power scheduling for remote state estimation," *Automatica*, vol. 50, pp. 1235–1242, Apr. 2014.
- [64] Y. Qi, P. Cheng, and J. Chen, "Optimal sensor data scheduling for remote estimation over a time-varying channel," *IEEE Trans. Autom. Control*, vol. 62, no. 9, pp. 4611–4617, Sep. 2017.
- [65] D. Han, Y. Mo, J. Wu, and L. Shi, "An opportunistic sensor scheduling solution to remote state estimation over multiple channels," *IEEE Trans. Signal Process.*, vol. 64, no. 18, pp. 4905–4917, Sep. 2016.
- [66] A. S. Leong, S. Dey, and D. E. Quevedo, "Sensor scheduling in variance based event triggered estimation with packet drops," *IEEE Trans. Autom. Control*, vol. 62, no. 4, pp. 1880–1895, Apr. 2017.
- [67] S. Liu, M. Fardad, E. Masazade, and P. K. Varshney, "Optimal periodic sensor scheduling in networks of dynamical systems," *IEEE Trans. Signal Process.*, vol. 62, no. 12, pp. 3055–3068, Jun. 2016.
- [68] C. Yang and L. Shi, "Deterministic sensor data scheduling under limited communication resource," *IEEE Trans. Signal Process.*, vol. 59, no. 10, pp. 5050–5056, Oct. 2011.
- [69] D. Han, J. Wu, Y. Mo, and L. Xie, "On stochastic sensor network scheduling for multiple processes," *IEEE Trans. Autom. Control*, vol. 62, no. 12, pp. 6633–6640, Dec. 2017.
- [70] L. Zhao, W. Zhang, J. Hu, A. Abate, and C. J. Tomlin, "On the optimal solutions of the infinite-horizon linear sensor scheduling problem," *IEEE Trans. Autom. Control*, vol. 59, no. 10, pp. 2825–2830, Oct. 2014.
- [71] M. Healy, T. Newe, and E. Lewis, "Wireless sensor node hardware: A review," in *Proc. IEEE Sensor*, Lecce, Italy, Oct. 2008, pp. 621–624.
- [72] J. L. Hill, and D. E. Culler, "Mica: A wireless platform for deeply embedded networks," *IEEE Micro*, vol. 22, no. 6, pp. 12–24, Nov./Dec. 2002.
- [73] M. Marin-Perianu *et al.*, "Decentralized enterprise systems: A multi-platform wireless sensor network approach," *IEEE Wireless Commun.*, vol. 14, no. 6, pp. 57–66, Dec. 2007.
- [74] P. Alriksson and A. Rantzer, "Experimental evaluation of a distributed Kalman filter algorithm," in *Proc. 46th IEEE Conf. Decis. Control*, New Orleans, LA, USA, Dec. 2007, pp. 5499–5504.
- [75] A.-S. Tonneau, N. Mitton, and J. Vandačle, "How to choose an experimentation platform for wireless sensor networks? A survey on static and mobile wireless sensor network experimentation facilities," *Ad Hoc Netw.*, vol. 30, pp. 115–127, 2015.
- [76] A. Kettner and M. Paolone, "Sequential discrete Kalman filter for real-time state estimation in power distribution systems: Theory and implementation," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 9, pp. 2358–2370, Sep. 2017.
- [77] G.-B. Dai, and Y.-C. Liu, "Distributed coordination and cooperation control for networked mobile manipulators," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5065–5074, Jun. 2017.
- [78] E. Nuño, I. Sarra, and L. Basañez, "Consensus in networks of nonidentical Euler–Lagrange systems using $P + d$ controllers," *IEEE Trans. Robot.*, vol. 29, no. 6, pp. 1503–1508, Dec. 2013.
- [79] D. Chen, Y. Zhang, and S. Li, "Tracking control of robot manipulators with unknown models: A Jacobian-matrix-adaption method," *IEEE Trans. Ind. Inform.*, vol. 14, no. 7, pp. 3044–3053, Jul. 2018.
- [80] T. Liu and Z.-P. Jiang, "Distributed formation control of nonholonomic mobile robots without global position measurements," *Automatica*, vol. 49, no. 2, pp. 592–600, Feb. 2013.

- [81] D. Chen and Y. Zhang, "Robust zeroing neural-dynamics and its time-varying disturbances suppression model applied to mobile robot manipulators," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4385–4397, Sep. 2018.
- [82] L. Tan, "Omnidirectional-vision-based distributed optimal tracking control for mobile multirobot systems with kinematic and dynamic disturbance rejection," *IEEE Trans. Ind. Electron.*, vol. 65, no. 7, pp. 5693–5703, Jul. 2018.
- [83] L. Jin, S. Li, L. Xiao, R. Lu, and B. Liao, "Cooperative motion generation in a distributed network of redundant robot manipulators with noises," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 48, no. 10, pp. 1715–1724, Oct. 2018.
- [84] Z. Zhang, Z. Yan, and T. Fu, "Varying-parameter RNN activated by finite-time functions for solving joint-drift problems of redundant robot manipulators," *IEEE Trans. Ind. Inform.*, vol. 14, no. 12, pp. 5359–5367, Dec. 2018, doi: 10.1109/TII.2018.2812757.
- [85] Z. Li and C.-Y. Su, "Neural-adaptive control of single-master-multiple-slaves teleoperation for coordinated multiple mobile manipulators with time-varying communication delays and input uncertainties," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 24, no. 9, pp. 1400–1413, Feb. 2013.
- [86] Y. Zhang, S. Li, and X. Zhou, "Recurrent neural network based velocity-level redundancy resolution for manipulators subject to joint acceleration limit," *IEEE Trans. Ind. Inform.*, vol. 66, no. 5, pp. 3573–3582, May 2019, doi: 10.1109/TIE.2018.2851960.
- [87] L. Jin, S. Li, X. Luo, Y. Li, and B. Qin, "Neural dynamics for cooperative control of redundant robot manipulators," *IEEE Trans. Ind. Inform.*, vol. 14, no. 9, pp. 3812–3821, Sep. 2018.
- [88] L. Jin, S. Li, H. La, and X. Luo, "Manipulability optimization of redundant manipulators using dynamic neural networks," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 4710–4720, Jun. 2017.
- [89] Z. Liu, C. Chen, Y. Zhang, and C. L. P. Chen, "Adaptive neural control for dual-arm coordination of humanoid robot with unknown nonlinearities in output mechanism," *IEEE Trans. Cybern.*, vol. 45, no. 3, pp. 521–532, Mar. 2015.
- [90] M. Wang and A. Yang, "Dynamic learning from adaptive neural control of robot manipulators with prescribed performance," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 8, pp. 2244–2255, Aug. 2017.
- [91] W. He, Z. Yan, Y. Sun, Y. Ou, and C. Sun, "Neural-learning-based control for a constrained robotic manipulator with flexible joints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 5993–6003, Dec. 2018, doi: 10.1109/TNNLS.2018.2803167.
- [92] W. He, B. Huang, Y. Dong, Z. Li, and C.-Y. Su, "Adaptive neural network control for robotic manipulators with unknown deadzone," *IEEE Trans. Cybern.*, vol. 48, no. 9, pp. 2670–2682, Sep. 2018.
- [93] Z. Li, Y. Xia, D. Wang, D.-H. Zhai, C.-Y. Su, and X. Zhao, "Neural network-based control of networked trilateral teleoperation with geometrically unknown constraints," *IEEE Trans. Cybern.*, vol. 46, no. 5, pp. 1051–1064, May 2016.
- [94] D.-H. Zhai, and Y. Xia, "Adaptive fuzzy control of multilateral asymmetric teleoperation for coordinated multiple mobile manipulators," *IEEE Trans. Fuzzy Syst.*, vol. 24, no. 1, pp. 57–70, Feb. 2016.
- [95] A. Marino, "Distributed adaptive control of networked cooperative mobile manipulators," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 5, pp. 1646–1660, Sep. 2018.
- [96] X. Ge and Q.-L. Han, "Distributed formation control of networked multi-agent systems using a dynamic event-triggered communication mechanism," *IEEE Trans. Ind. Electron.*, vol. 64, no. 10, pp. 8118–8127, Oct. 2017.
- [97] W. He, B. Zhang, Q.-L. Han, and F. Qian, "Leader-following consensus of nonlinear multiagent systems with stochastic sampling," *IEEE Trans. Cybern.*, vol. 47, no. 2, pp. 327–338, Feb. 2017.
- [98] L. Ding, Q.-L. Han, X. Ge, and X.-M. Zhang, "An overview of recent advances in event-triggered consensus of multiagent systems," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1110–1123, Apr. 2018.
- [99] X. Ge, Q.-L. Han, D. Ding, X.-M. Zhang, and B. Ning, "A survey on recent advances in distributed sampled-data cooperative control of multi-agent systems," *Neurocomputing*, vol. 275, pp. 1684–1701, Jan. 2018.
- [100] P. N. Köhler, M. A. Müller, and F. Allgöwer, "A distributed economic MPC framework for cooperative control under conflicting objectives," *Automatica*, vol. 96, pp. 368–379, Oct. 2018.
- [101] E. Asadi, and A. Richards, "Scalable distributed model predictive control for constrained systems," *Automatica*, vol. 93, pp. 407–414, Jul. 2018.
- [102] I. Necoara, and J. A. Suykens, "Application of a smoothing technique to decomposition in convex optimization," *IEEE Trans. Autom. Control*, vol. 53, no. 11, pp. 2674–2679, Dec. 2008.
- [103] Z. Wang and C.-J. Ong, "Accelerated distributed MPC of linear discrete-time systems with coupled constraints," *IEEE Trans. Autom. Control*, vol. 63, no. 11, pp. 3838–3849, Nov. 2018.
- [104] P. Giselsson, M. D. Doan, T. Keviczky, B. D. Schutter, and A. Rantzer, "Accelerated gradient methods and dual decomposition in distributed model predictive control," *Automatica*, vol. 49, no. 3, pp. 829–833, Mar. 2013.
- [105] L. Dai, Y. Xia, Y. Gao, and M. Cannon, "Distributed stochastic MPC of linear systems with additive uncertainty and coupled probabilistic constraints," *IEEE Trans. Autom. Control*, vol. 62, no. 7, pp. 3474–3481, Jul. 2017.
- [106] S. Cominesi, M. Farina, L. Giulioni, B. Picasso, and R. Scattolini, "A two-layer stochastic model predictive control scheme for microgrids," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 1, pp. 1–13, Jan. 2018.
- [107] M. Farina and R. Scattolini, "Distributed predictive control: A non-cooperative algorithm with neighbor-to-neighbor communication for linear systems," *Automatica*, vol. 48, no. 6, pp. 1088–1096, Jun. 2012.
- [108] S. Rivero and G. Ferrari-Trecate, "Tube-based distributed control of linear constrained systems," *Automatica*, vol. 48, no. 11, pp. 2860–2865, Nov. 2012.
- [109] S. Lucia, M. Kögel, and R. Findeisen, "Contract-based predictive control of distributed systems with plug and play capabilities," *IFAC-PapersOnLine*, vol. 48, no. 23, pp. 205–211, 2015.
- [110] P. A. Trodden, and J. M. Maestre, "Distributed predictive control with minimization of mutual disturbances," *Automatica*, vol. 77, pp. 31–43, Mar. 2017.
- [111] P. Liu and U. Ozguner, "Distributed model predictive control of spatially interconnected systems using switched cost functions," *IEEE Trans. Autom. Control*, vol. 63, no. 7, pp. 2161–2167, Jul. 2018.
- [112] F. Berkel and S. Liu, "An event-triggered output-based model predictive control strategy," *IEEE Trans. Control Netw. Syst.*, to be published, doi: 10.1109/TCNS.2018.2878506.
- [113] S. Roshany-Yamchi *et al.*, "Kalman filter-based distributed predictive control of large-scale multi-rate systems: Application to power networks," *IEEE Trans. Control Syst. Technol.*, vol. 21, no. 1, pp. 27–39, Jan. 2013.
- [114] P. Giselsson and A. Rantzer, "On feasibility, stability and performance in distributed model predictive control," *IEEE Trans. Autom. Control*, vol. 59, no. 4, pp. 1031–1036, Apr. 2014.
- [115] K. Meng, Z. Dong, Z. Xu, and S. R. Weller, "Cooperation-driven distributed model predictive control for energy storage systems," *IEEE Trans. Smart Grid*, vol. 6, no. 6, pp. 2583–2585, Nov. 2015.
- [116] H. Li and Y. Shi, "Robust distributed model predictive control of constrained continuous-time nonlinear systems: A robustness constraint approach," *IEEE Trans. Autom. Control*, vol. 59, no. 6, pp. 1673–1678, Jun. 2014.
- [117] L. Ding, Q.-L. Han, L. Wang, and E. Sindi, "Distributed cooperative optimal control of DC microgrids with communication delays," *IEEE Trans. Ind. Inform.*, vol. 14, no. 9, pp. 3924–3935, Sep. 2018.
- [118] L. Ding, Q.-L. Han, and X.-M. Zhang, "Distributed secondary control for active power sharing and frequency regulation in islanded microgrids using an event-triggered communication mechanism," *IEEE Trans. Ind. Inform.*, to be published, doi: 10.1109/TII.2018.2884494.
- [119] N. Dehkordi, N. Sadati, and M. Hamzeh, "Fully distributed cooperative secondary frequency and voltage control of islanded microgrids," *IEEE Trans. Energy Convers.*, vol. 32, no. 2, pp. 675–685, Jun. 2017.
- [120] N. Dehkordi, N. Sadati, and M. Hamzeh, "Distributed robust finite-time secondary voltage and frequency control of islanded microgrids," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3648–3659, Sep. 2017.
- [121] A. Bidram, A. Davoudi, F. Lewis, and J. Guerrero, "Distributed cooperative secondary control of microgrids using feedback linearization," *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 3462–3470, Aug. 2013.
- [122] A. Bidram, F. L. Lewis, and A. Davoudi, "Distributed control systems for small-scale power networks: Using multiagent cooperative control theory," *IEEE Control Syst. Mag.*, vol. 34, no. 6, pp. 56–77, Dec. 2014.
- [123] X. Wu, C. Shen, and R. Iravani, "A distributed, cooperative frequency and voltage control for microgrids," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 2764–2776, Jul. 2018.
- [124] X. Lu, X. Yu, J. Lai, J. M. Guerrero, and H. Zhou, "Distributed secondary voltage and frequency control for islanded microgrids with uncertain communication links," *IEEE Trans. Ind. Inform.*, vol. 13, no. 2, pp. 448–460, Apr. 2017.

- [125] S. Abhinav, I. D. Schizas, F. L. Lewis, and A. Davoudi, "Distributed noise-resilient networked synchrony of active distribution systems," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 836–846, Mar. 2018.
- [126] S. Manaffam, M. Talebi, A. K. Jainy, and A. Behal, "Intelligent pinning based cooperative secondary control of distributed generators for microgrid in islanding operation mode," *IEEE Trans. Power Syst.*, vol. 33, no. 2, pp. 1364–1373, Feb. 2018.
- [127] M. E. Romero and M. M. Seron, "Ultimate boundedness of voltage droop control with distributed secondary control loops," *IEEE Trans. Smart Grid*, to be published, doi: 10.1109/TSG.2018.2849583.
- [128] J. W. Simpson-Porco, Q. Shafiee, F. Dörfler, J. C. Vasquez, J. M. Guerrero, and F. Bullo, "Secondary frequency and voltage control of islanded microgrids via distributed averaging," *IEEE Trans. Ind. Electron.*, vol. 62, no. 11, pp. 7025–7038, Nov. 2015.
- [129] Q. Shafiee, J. M. Guerrero, and J. C. Vasquez, "Distributed secondary control for islanded microgrids—A novel approach," *IEEE Trans. Power Electron.*, vol. 29, no. 2, pp. 1018–1031, Feb. 2014.
- [130] R. Heidari, M. M. Seron, and J. H. Braslavsky, "Ultimate boundedness and regions of attraction of frequency droop controlled microgrids with secondary control loops," *Automatica*, vol. 81, pp. 416–428, Jul. 2017.
- [131] J. W. Simpson-Porco, F. Dörfler, and F. Bullo, "Synchronization and power sharing for droop-controlled inverters in islanded microgrids," *Automatica*, vol. 49, no. 9, pp. 2603–2611, Sep. 2013.
- [132] F. Dörfler, J. W. Simpson-Porco, and F. Bullo, "Breaking the hierarchy: distributed control and economic optimality in microgrids," *IEEE Trans. Control Netw. Syst.*, vol. 3, no. 3, pp. 241–253, Sep. 2016.
- [133] C.-Y. Chang and W. Zhang, "Distributed control of inverter-based lossy microgrids for power sharing and frequency regulation under voltage constraints," *Automatica*, vol. 66, pp. 85–95, Apr. 2016.
- [134] V. Singh, N. Kishor, and P. Samuel, "Distributed multi-agent system-based load frequency control for multi-area power system in smart grid," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5151–5160, Jun. 2017.
- [135] C.-K. Zhang, L. Jiang, Q. H. Wu, Y. He, and M. Wu, "Further results on delay-dependent stability of multi-area load frequency control," *IEEE Trans. Power Syst.*, vol. 28, no. 4, pp. 4465–4474, Nov. 2013.
- [136] D. Pullaguram, S. Mishra, and N. Senroy, "Event-triggered communication based distributed control scheme for DC microgrid," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 5583–5593, May 2018.
- [137] S. Sahoo and S. Mishra, "An adaptive event-triggered communication based distributed secondary control for DC microgrids," *IEEE Trans. Smart Grid*, vol. 9, no. 6, pp. 6674–6683, Jun. 2018.
- [138] M. S. Golsorkhi, D. J. Hill, and H. R. Karshenas, "Distributed voltage control and power management of networked microgrids," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 6, no. 4, pp. 1892–1902, Dec. 2018, doi: 10.1109/JESTPE.2017.2773138.
- [139] J. Qin, M. Li, L. Shi, and X. Yu, "Optimal denial-of-service attack scheduling with energy constraint over packet-dropping networks," *IEEE Trans. Autom. Control*, vol. 63, no. 6, pp. 1648–1663, Jun. 2018.
- [140] W. Zeng, Y. Zhang, and M.-Y. Chow, "Resilient distributed energy management subject to unexpected misbehaving generation units," *IEEE Trans. Ind. Inform.*, vol. 13, no. 1, pp. 208–216, Feb. 2017.
- [141] Y. Feng, B. Hu, H. Hao, Y. Gao, Z. Li, and J. Tan, "Design of distributed cyber-physical systems for connected and automated vehicles with implementing methodologies," *IEEE Trans. Ind. Inform.*, vol. 14, no. 9, pp. 4200–4211, Sep. 2018.
- [142] J. Zhang, R. S. Blum, X. Lu, and D. Conus, "Asymptotically optimum distributed estimation in the presence of attacks," *IEEE Trans. Signal Process.*, vol. 63, no. 5, pp. 1086–1101, Mar. 2015.
- [143] R. Deng, G. Xiao, and R. Lu, "Defending against false data injection attacks on power system state estimation," *IEEE Trans. Ind. Inform.*, vol. 13, no. 1, pp. 198–207, Feb. 2017.
- [144] Q. Yang, J. Yang, W. Yu, D. An, N. Zhang, and W. Zhao, "On false data-injection attacks against power system state estimation: Modeling and countermeasures," *IEEE Trans. Parallel Distrib. Syst.*, vol. 25, no. 3, pp. 717–729, Mar. 2014.
- [145] M. N. Kurt, Y. Yilmaz, and X. Wang, "Distributed quickest detection of cyber-attacks in smart grid," *IEEE Trans. Inf. Forensics Secur.*, vol. 13, no. 8, pp. 2015–2030, Aug. 2018.
- [146] S. Chen, X. Chen, Z. Pei, X. Zhang, and H. Fang, "Distributed filtering algorithm based on tunable weights under untrustworthy dynamics," *IEEE/CAA J. Autom. Sin.*, vol. 3, no. 2, pp. 225–232, Apr. 2016.
- [147] C. Liang, F. Wen, and Z. Wang, "Trust-based distributed Kalman filtering for target tracking under malicious cyber attacks," *Inf. Fusion*, vol. 46, pp. 44–50, Mar. 2019.
- [148] Y. Liu, H. Xin, Z. Qu, and D. Gan, "An attack-resilient cooperative control strategy of multiple distributed generators in distribution networks," *IEEE Trans. Smart Grid*, vol. 7, no. 6, pp. 2923–2932, Nov. 2016.
- [149] M. Zhu and S. Martínez, "On distributed constrained formation control in operator-vehicle adversarial networks," *Automatica*, vol. 49, no. 12, pp. 3571–3582, Dec. 2013.
- [150] J. Duan, W. Zeng, and M.-Y. Chow, "Resilient distributed DC optimal power flow against data integrity attack," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3543–3552, Jul. 2018.



Derui Ding (M'16) received the B.Sc. degree in industry engineering and the M.Sc. degree in detection technology and automation equipment from Anhui Polytechnic University, Wuhu, China, in 2004 and 2007, respectively, and the Ph.D. degree in control theory and control engineering from Donghua University, Shanghai, China, in 2014.

From July 2007 to December 2014, he was a Teaching Assistant and then a Lecturer with the Department of Mathematics, Anhui Polytechnic University. He is currently a Senior Research Fellow with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC, Australia. From June 2012 to September 2012, he was a Research Assistant with the Department of Mechanical Engineering, The University of Hong Kong, Hong Kong. From March 2013 to March 2014, he was a visiting scholar with the Department of Information Systems and Computing, Brunel University London, Uxbridge, U.K. He authored or coauthored more than 40 papers in refereed international journals. His research interests include nonlinear stochastic control and filtering, as well as multi-agent systems and sensor networks.

Dr. Ding is an Associate Editor for *Neurocomputing*. He is also a very active reviewer for many international journals.



Qing-Long Han (M'09–SM'13–F'19) received the B.Sc. degree in mathematics from Shandong Normal University, Jinan, China, in 1983, and the M.Sc. and Ph.D. degrees in control engineering and electrical engineering from East China University of Science and Technology, Shanghai, China, in 1992 and 1997, respectively.

From September 1997 to December 1998, he was a Postdoctoral Researcher Fellow with the Laboratoire d'Automatique et d'Informatique Industrielle (currently, Laboratoire d'Informatique et d'Automatique pour les Systèmes), École Supérieure d'Ingénieurs de Poitiers (currently, École Nationale Supérieure d'Ingénieurs de Poitiers), Université de Poitiers, Poitiers, France. From January 1999 to August 2001, he was a Research Assistant Professor with the Department of Mechanical and Industrial Engineering, Southern Illinois University at Edwardsville, Edwardsville, IL, USA. From September 2001 to December 2014, he was a Laureate Professor, an Associate Dean (Research and Innovation) with the Higher Education Division, and the Founding Director of the Centre for Intelligent and Networked Systems with Central Queensland University, Australia. From December 2014 to May 2016, he was the Deputy Dean (Research) with the Griffith Sciences and a Professor with the Griffith School of Engineering, Griffith University, Brisbane, QLD, Australia. In May 2016, he joined Swinburne University of Technology, Melbourne, VIC, Australia, where he is currently the Pro-Vice-Chancellor (Research Quality) and a Distinguished Professor. In March 2010, he was appointed the Chang Jiang (Yangtze River) Scholar Chair Professor by Ministry of Education, China. His research interests include networked control systems, multi-agent systems, time-delay systems, complex dynamical systems, and neural networks.

Prof. Han is a Fellow of The Institution of Engineers Australia. He is an Associate Editor for several international journals, including the IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE INDUSTRIAL ELECTRONICS MAGAZINE, IEEE/CAA JOURNAL OF AUTOMATICA SINICA, *Control Engineering Practice*, and *Information Sciences*. He is one of The World's Most Influential Scientific Minds: 2014–2016 and 2018. He is a Highly Cited Researcher according to Clarivate Analytics (formerly Thomson Reuters).



Zidong Wang (SM'03–F'14) was born in Jiangsu, China, in 1966. He received the B.Sc. degree in mathematics from Suzhou University, Suzhou, China, in 1986, and the M.Sc. degree in applied mathematics and the Ph.D. degree in electrical engineering from Nanjing University of Science and Technology, Nanjing, China, in 1990 and 1994, respectively.

He is currently a Professor of Dynamical Systems and Computing with the Department of Computer Science, Brunel University London, Uxbridge, U.K. From 1990 to 2002, he held teaching and research appointments in universities in China, Germany, and the U.K. He has authored or coauthored more than 400 papers in refereed international journals. His research interests include dynamical systems, signal processing, bioinformatics, and control theory and applications.

Prof. Wang is a Fellow of the Royal Statistical Society and a member of program committee for many international conferences. He serves (or has served) as the Editor-in-Chief for *Neurocomputing*, the Deputy Editor-in-Chief for *International Journal of Systems Science*, and an Associate Editor for 12 international journals including the IEEE TRANSACTIONS ON AUTOMATIC CONTROL, IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY, IEEE TRANSACTIONS ON NEURAL NETWORKS, IEEE TRANSACTIONS ON SIGNAL PROCESSING, and IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS, PART C. He was the recipient of the Alexander von Humboldt Research Fellowship of Germany, the JSPS Research Fellowship of Japan, and William Mong Visiting Research Fellowship of Hong Kong.



Xiaohua Ge (M'18) received the B.Eng. degree in electronics and information engineering from Nanchang Hangkong University, Nanchang, China, in 2008, the M.Eng. degree in control theory and control engineering from Hangzhou Dianzi University, Hangzhou, China, in 2011, and the Ph.D. degree in computer engineering from Central Queensland University, Rockhampton, QLD, Australia, in 2014.

From 2011 to 2013, he was a Research Assistant with the Centre for Intelligent and Networked Systems (CINS, now known as Centre for Intelligent Systems), Central Queensland University, Rockhampton, QLD, Australia. In 2014, he was a Research Fellow with the CINS, Central Queensland University. From 2015 to 2017, he was a Research Fellow with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. He is currently a Lecturer with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC, Australia. His research interests include distributed estimation over sensor networks, distributed coordination in multi-agent systems, security and privacy preserving in cyber-physical systems.